

**A MINI PROJECT REPORT**  
**ON**  
**Empowering Snort with OpenVAS Scanning and Splunk**  
**Visualization**

**BACHELOR OF TECHNOLOGY**  
**In**  
**Computer Science & Engineering (AI)**  
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## **Certificate**

This is to certify that the Mini Project report entitled “**Empowering SNORT with OPENVAS Vulnerability Scanner and Splunk Visualization**” done by **Anshu Kumar (2301921529002)**, **Hemant Sharma (2301921529006)**, **Husain jamil Khan (2301921529007)** and **Vidit (2301921529012)** is an original work carried out by them in Department of Computer Science and Engineering (AI), G.L. Bajaj Institute of Technology & Management, Greater Noida under my supervision. The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

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## ABSTRACT

In today's interconnected world, a comprehensive cybersecurity solution is essential. This project aims to improve network security by integrating Snort IDS/IPS, Splunk, and OpenVAS Vulnerability Scanner in an innovative manner. By combining these open-source tools, the research aims to provide a comprehensive solution for identifying network system vulnerabilities and potential threats. In a controlled environment, the research is conducted using Kali Linux, Metasploitable, and cloud-hosted Splunk. The method includes data collection, parsing OpenVAS reports, converting to a Snort-compatible format, automating rule creation with Python scripts, and integrating with Splunk for real-time visualization in order to improve the performance of rule-based intrusion detection systems and generate intrusion alerts. In addition, Splunk is required to visualize intrusions and anomalies detected by Snort. This project's primary objective is to address the current challenges of delayed threat detection, accurate threat visualization, and manual rule formulation. It intends to bridge the gap between vulnerability assessment and real-time threat detection, augmenting network defences comprehensively and effectively. This integration aims to contribute to the continuous evolution of intrusion detection and network security practices.

**Keywords:** Intrusion Detection System, Network Security, Snort, OpenVAS, Splunk, Vulnerability Assessment, Cyber Security, Log Management, Rule-based IDS, Network Threats

## Preface

The present project is an outcome arising from the integration of innovation, cybersecurity, and viability. In the interconnected world and with the digitization of various processes, safeguarding sensitive data and upholding secure network environments is of the highest priority. The concept presented within this research centres on the progression, combination, and interrelation of three innovative network security technologies, namely Snort, OpenVAS, and Splunk.

Considering the rapid growth of threats and the continual expansion of potential targets, the incorporation of intrusion detection and vulnerability assessment techniques has become progressively indispensable. This project examines the technical complexity involved in integrating many components, as well as delves into the ethical, legal, and social implications associated with such integration.

This project emphasizes the significance of acquiring a comprehensive comprehension of the selected approach, the challenges that were effectively overcome, and the significant insights that were acquired.

The integration of Snort, OpenVAS, and Splunk serves as a tangible demonstration of using technology to implement proactive protection measures within a globally networked setting. The automation of intrusion detection rule creation enables us to effectively address the challenges posed by a constantly evolving threat environment, hence enhancing the efficiency of rule management.

This undertaking would have been impossible without the guidance of mentors, the support of colleagues, and the determination of individuals who contributed to the validation of this research.

As the study advances through subsequent sections, delving into detailed technical aspects, examining findings, and considering future ramifications, the aspiration is for this project to offer a modest yet noteworthy contribution to the domain of cybersecurity. This initiative aims to showcase both technical expertise and the innovative and resilient nature that defines the cybersecurity field, with each keystroke.

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# **Chapter 1: Introduction**

In an era defined by digital transformation and interconnectedness, the paramount importance of network security is undeniable. Attackers are everywhere, watching our digital assets closely. Malicious actors are developing increasingly sophisticated malicious software and tools in order to exploit existing system vulnerabilities. We require a robust cybersecurity system to defend against cyberattacks and intrusions.

The expanding number of networked devices, together with the constant flow of information, causes tough challenges in ensuring data security, availability, and integrity. Prioritizing network security is critical for overcoming these issues.

Security solutions like intrusion detection systems (IDS) and Vulnerability scanners play a key role in preventing unauthorized access, data breaches, and potential threats. This project embarks on a journey to explore the integration of three pivotal tools – Snort, OpenVAS, and Splunk – to enhance intrusion detection capabilities and streamline vulnerability assessment.

## **1.1. Background and Context**

As Organizations heavily depend on digital infrastructure, the risk of cyber threats looms larger than ever. In the age of network security, Intrusion Detection Systems (IDS) play a vital role by effectively detecting and promptly addressing incidents of unauthorized access, anomalies, and malicious actions. Additionally, Vulnerability Assessment Tools like OpenVAS help in the identification of vulnerabilities and weaknesses of a system or network. The individual significance of these tools is noteworthy; nevertheless, their combination offers the potential for a more comprehensive and streamlined approach to network security.

## **1.2. Problem Statement**

While the individual benefits of Snort, OpenVAS, and Splunk are acknowledged, there is a lack of research into their collective utility. The difficulty of efficiently producing intrusion detection rules based on identified vulnerabilities is a common obstacle for businesses. Creating criteria manually for each vulnerability can be difficult, time-consuming, and error-prone. This project addresses this issue by proposing the integration of these tools to automate rule generation.

### **1.3. Research Objectives**

The primary goal of this study is to enhance network security by integrating Snort, OpenVAS, and Splunk. Among the specific objectives are:

1. Integrating Snort and OpenVAS to expedite the creation of intrusion detection rules.
2. Developing Python scripts to automate the conversion of OpenVAS reports to a Snort compatible format.
3. Evaluate the capability of the integrated system to detect hazards and vulnerabilities.
4. The moral, legal, and social implications of automated intrusion detection rule generation are investigated.

### **1.4. Scope and Limitations**

The research is conducted in a controlled environment using Kali Linux and Metasploitable to simulate real-world network interactions and potential attacks. While the integration of Snort, OpenVAS, and Splunk has the potential for diverse network configurations, this project focuses predominantly on demonstrating the approach's viability and efficiency in this particular environment.

### **1.5. Significance of the Study**

Integrating Snort, OpenVAS, and Splunk provides a multidimensional approach to network security. Automated rule generation contributes to proactive defence, allowing organizations to respond promptly to potential threats. In addition, the study explores the ethical and legal dimensions of network security automation, which reflects the broader societal impact of technological innovations.

## **1.6. Overview**

The subsequent segments of this project delve into the specifics of each aspect. The second chapter provides a comprehensive literature review that lays the groundwork for comprehending the significance of intrusion detection, vulnerability assessment, and log management. The third chapter describes the methodology used to accomplish integration, automate rule generation, and ethical considerations. The subsequent chapters describe the implementation process, the analysis of results, and the implications and future prospects.

## **1.7. Roadmap**

The road map for this project includes integration, automation, and analysis. The subsequent sections investigate the complexities of this integration and cast light on its technique, implementation, results, and repercussions. Chapter 2 contains a comprehensive literature review that follows this introduction. A new section analyses the integrated system in Chapter 3.

This section investigates legal, social, ethical, and professional factors. Chapter 4 describes integration methodologies for Snort, OpenVAS, and Splunk. The fifth chapter describes the practical implementation of the integrated system, while the sixth chapter provides the results and insightful analysis. In Chapter 7, repercussions, consequences, and broader analyses of these findings are presented, leading to Chapter 8's conclusion. The bibliography and any other appendices are located at the end of the report.

## Chapter 2: Technological Foundations: Snort, OpenVAS, and Splunk

### 2.1 Intrusion Detection and Prevention Systems: A Foundation in Cybersecurity

Intrusion detection and prevention systems (IDS/IPS) founded on cybersecurity are essential for protecting networks and systems in contemporary digital environments. In terms of network security, this section investigates the functions, distinctions, and significance of IDS and IPS.

#### Intrusion Detection System (IDS):

In the constantly changing cybersecurity industry, it is crucial to prevent unauthorized access to digital assets by malicious actors. Intrusion Detection Systems (IDS) acquire control in this scenario. IDS analyses system activity and network traffic for unauthorized access, hostile behaviour, and abnormalities that could compromise a digital environment's security.

#### Understanding IDS Types:

1. **Signature-Based IDS:** It uses a large database of predefined signatures, patterns, and rules and is also known as a rule-based IDS. These rules have been carefully created to identify typical attack methods. Whenever network traffic matches these signatures, an alert is generated. Signature-based IDSs excel at identifying well-known threats but might be unable to recognize new attacks that lack predefined signatures.
2. **Anomaly-Based IDS:** Anomaly-based IDS, unlike its counterpart, establishes a baseline for normal network behaviour. Any deviation from this established norm is identified as a possible threat. Anomaly-based IDS excels at detecting new or "zero-day" attacks, but the possibility of false positives due to legitimate network behaviour variations occurs.

### **Intrusion Prevention System (IPS):**

IPS not only identifies but also rapidly eliminates any potential threats. Unlike IDS, which alerts administrators to suspected intrusions, IPS quickly prevents or mitigates recognized risks. This capability transforms IPS into a proactive network guardian, detecting and defending against threats before they cause harm.

#### **There are numerous steps involved in connecting IDS and IPS in practice:**

1. **Monitoring:** A continuous monitor is kept on network traffic, carefully inspecting packets and data streams for any anomalies, patterns, or actions that fit certain criteria.
2. **Detection:** The system compares the observed behaviour to baseline patterns, rules, or signatures when a potential threat is discovered. When there is a match, an alert is generated.
3. **Alert Generation:** Alerts are the alarm sounds, warning security analysts or administrators about the potential threat. The nature and severity of the identified behaviour are elaborated upon in these alerts, which offer complete information.
4. **IPS Response:** IPS responds instantly to threats due to its real-time reaction capabilities. By blocking malicious traffic, changing network settings, or isolating impacted systems, it stops the threat from spreading.
5. **Analysis and Reporting:** Security teams focus on alert analysis after the initial response. The incident is examined, and essential information is acquired for further investigation. This key phase clarifies the threat's nature and directs the development of preventive actions against future occurrences of similar situations.

## Unveiling Snort: Uniting Power and Flexibility

### Introduction to Snort:

Snort is an open-source intrusion detection and prevention system pioneer. Martin Roesch's innovative thinking was critical in the late 1990s creation of Snort, and its adaptability and opensource nature influenced the cybersecurity domain. The capacity to detect a wide range of network-based threats and vulnerabilities is its key advantage.

### Components of Snort: A Symphony of Defence:

1. **Packet Sniffer:** Snort takes on the function of a packet sniffer, adeptly intercepting network packets and inspecting their headers for indicators of alarming patterns.
2. **Packet Logger:** Snort in logger mode systematically logs detected events on the system, allowing administrators to go through and analyse scenarios in the past.
3. **Network Intrusion Detection Mode:** Snort, as a standard IDS, methodically analyses network packets against a large database of predetermined signatures and patterns.
4. **Network Intrusion Prevention Mode:** Using a proactive approach, this mode goes beyond conventional alerts, actively preventing malicious traffic before it causes damage.
5. **Rule Engine:** The rule engine is at the core of Snort's detection ability. Packets are compared to a set of rules that identify distinct patterns, behaviours, or characteristics associated with various attack types.
6. **Pre-processors:** Pre-processors establish the framework for analysis by preparing packets. The accuracy of analysis is improved by reassembling fragmented packets and normalizing data.
7. **Rule Options:** Rule options provide context to the rule engine by defining the criteria and behaviours that determine whether an alert should be triggered.



## The Art of Creating Snort Rules:

Snort's effectiveness is dependent on its rule-based methodology. These rules, which are a combination of properties and behaviours, are skilfully defined using a specialized syntax. The artistry is found in:

1. **Rule Header:** An action, protocol, source IP address, source port, direction operator, destination IP address, and destination port coordinated composition.
2. **Rule Options:** The orchestration extends to rule options, providing detailed context to detection by specifying qualities, patterns, or behaviours associated with the threat.
3. **Payload:** In the payload part, Snort searches for content patterns to match against network traffic, highlighting potential dangers.
4. **Additional Options:** The composition builds up with more possibilities, allowing the rule's behaviour to be refined. The symphony is enhanced by flow direction, content modifiers, and metadata.

## The Power and Versatility of Snort:

Snort is a landmark among Intrusion Detection and Prevention Systems (IDS/IPS) due to its attractive blend of power and versatility. Snort's open-source nature provides organizations with the unique opportunity to precisely adjust its capabilities to their needs, ensuring a responsive defence against the ever-changing cyber world.

The global cybersecurity community collaborated to create the symphony of Snort's capabilities, which resonates across its large rule library. This repository contains a wide range of prebuilt rules, each designed to identify different forms of attacks. These guidelines, when combined, build a strong defence that adjusts to new threats.

Snort's canvas, on the other hand, goes beyond the predefined. Organizations can develop customized rules to improve their defence methods and anticipate new attack vectors. This customization enables organizations to remain adaptable and proactive in dealing with cyber threats.

Snort's ease of use bridges the gap between cybersecurity specialists and beginners. Its user-friendly design guarantees that a broader range of practitioners can leverage its capabilities, making cybersecurity expertise more accessible.

## **2.2 Vulnerability Scanners and OpenVAS: Navigating Digital Weaknesses**

In the ever-changing landscape of cybersecurity, identifying and mitigating vulnerabilities are key components for fortifying digital environments against possible attackers. This section looks into the field of vulnerability scanners, giving light on their significance, and features, and, in particular, focuses on OpenVAS as a prominent example in this market.

### **Understanding Vulnerability Scanners:**

To address the potential threat of unauthorized access by fraudulent individuals to digital systems, apps, and networks, it is imperative to employ vulnerability scanners. Instances of vulnerabilities encompass unsecured configurations, obsolete software, and unpatched issues about operating systems and applications.

Various forms of vulnerability scanning, such as active and passive monitoring, serve to fulfil this pivotal function. Active scanning is a technique employed to identify vulnerabilities by rapidly establishing a network connection with the designated target system. Conversely, passive evaluations entail the analysis of network data to detect possible vulnerabilities without causing any system interruption.

### **Introducing OpenVAS:**

The capabilities of this technology area are best illustrated by the well-known OpenVAS, or Open Vulnerability Assessment System, vulnerability scanner. For its extensive methodology, Greenbone Networks' OpenVAS vulnerability assessment tool is well-known in the industry. It is well regarded for its ability to give precise results, adaptability in many contexts, and strong community support.

## Features of OpenVAS:

1. **Network Scanning:** OpenVAS conducts thorough network scans, looking for known vulnerabilities in hosts, services, and applications. Organizations can thoroughly discover potential entry points for attackers using this technique.
2. **Regular Updates:** Any vulnerability scanner's success depends on its capacity to keep up with the always-changing threat landscape. In this regard, OpenVAS stands out, maintaining a current database of known vulnerabilities to provide precise identification.
3. **Customization:** Organizations can customize their scans with OpenVAS to meet their own needs. They can focus on important systems, particular vulnerabilities, or particular services due to this personalization.
4. **Comprehensive Reporting:** The results of OpenVAS scans are provided in thorough reports. These reports highlight the vulnerabilities that have been found, their seriousness, and the suggested fixes.
5. **Integration with Other Tools:** To improve its usability and expand its capabilities within an organization's current security architecture, OpenVAS interacts effortlessly with other cybersecurity products and platforms.

## OpenVAS Workflow:

The OpenVAS workflow is a systematic process designed to identify, assess, and categorize vulnerabilities:

1. **Target Selection:** Selecting the targets for vulnerability assessment is the first step. This might apply to both specific hosts and entire networks.
2. **Scanning:** The scan is started by OpenVAS, which uses a variety of ways to uncover potential vulnerabilities. To match against the target, it uses its huge database of known vulnerabilities.
3. **Detection and Analysis:** The severity of the vulnerability and its possible impact on the system are carefully examined. The remedial actions will be prioritized using this analysis.

4. **Reporting:** The scan results are then put into thorough reports that describe the vulnerabilities, their risks, and practical advice for resolving them.
5. **Remediation:** Organizations can resolve the identified vulnerabilities using the information provided by OpenVAS. This could include patching, modifying settings, or upgrading software.

## **Conclusion:**

In the ongoing fight against cyber intrusions, OpenVAS, and other vulnerability scanners are indispensable. Their ability to identify and prioritize vulnerabilities empowers organizations with the knowledge necessary to effectively defend digital landscapes. OpenVAS demonstrates through its extensive features, open-source concept, and active community how technology can assist organizations in managing the complexities of cybersecurity, strengthening their defences, and protecting their digital assets from attackers.

## **2.3 Splunk: Illuminating Insights through Data Analytics**

In the ever-changing field of cybersecurity, where threats evolve and attacks become more sophisticated, comprehensive and real-time insights are of the utmost importance. This section examines Splunk, a market-leading data analytics platform that enables organizations to leverage the power of their data, discover meaningful patterns, and defend themselves proactively from cyber threats.

Splunk is a platform for data analysis and visualization that extracts meaningful insights from unstructured data. Splunk's ability to collect, index, and correlate data from structured and unstructured sources gives businesses a comprehensive view that allows them to discover anomalies respond quickly, and make smart decisions.

### **Key Features and Capabilities:**

1. **Data Collection:** Splunk takes data from a variety of sources, including logs, metrics, and events from applications, networks, and systems. This extensive strategy allows for a thorough awareness of an organization's digital landscape.
2. **Real-Time Search and Analysis:** Splunk's real-time search and analysis engine allows businesses to analyze data in real-time, finding trends, patterns, and anomalies as they emerge. This feature enables immediate response to developing threats.
3. **Correlation and Visualization:** Splunk excels at bridging the gaps between diverse data sources. It uncovers hidden connections by correlating seemingly unrelated events and presents them through interactive visualizations and dashboards.
4. **Machine Learning and AI:** Splunk incorporates machine learning and artificial intelligence capabilities to enhance threat detection. It facilitates proactive defence by identifying deviations from established norms via predictive analytics and anomaly detection.
5. **Alerting and Monitoring:** Splunk provides alerting mechanisms that can be configured to send notifications based on predefined thresholds or patterns. This real-time notification ensures a prompt response to any potential security incidents.
6. **Forensics and Investigation:** Splunk's comprehensive data records facilitate thorough investigations following an incident. Analysts can retrace events, comprehend the action chain, and evaluate the scope and impact of an attack.

## **Splunk Workflow:**

Splunk's effectiveness stems from its well-defined workflow:

1. **Data Collection:** Splunk collects data from multiple sources, such as logs, APIs, and external systems.
2. **Indexing and Parsing:** The ingested data is processed and indexed for easy search and analysis.
3. **Search and Analysis:** Splunk's powerful search language is used by analysts to examine data, find patterns, and spot anomalies.
4. **Visualization:** Insights from data analysis are presented through interactive visualizations and dashboards, which makes complex information easier to interpret.
5. **Alerting and Automation:** Splunk generates alerts based on predefined criteria, allowing for quick response to potential risks.
6. **Investigation and Remediation:** Splunk facilitates thorough investigation in the event of a threat, assisting in determining the nature and scope of the issue.

## **Conclusion:**

Splunk transcends its data analysis function to become a significant cybersecurity tool. It enables organizations to respond with agility and precision to evolving threats by supplying them with real-time insights, robust analytics, and proactive threat detection.

Splunk's contribution to increasing cyber resilience is based not only on its technological expertise, but also on its ability to enable cybersecurity teams to make informed decisions, strengthen defences, and navigate the complex landscape of digital security with assurance.

## Chapter 3: Literature Review

Integrating Snort, OpenVAS, and Splunk within the context of network security is an important step toward fortifying digital environments against emergent cyber threats. This chapter conducts a thorough literature review to determine the significance, difficulties, and opportunities of this integration.

This section analyses significant research studies, methodologies, and findings to describe the landscape of intrusion detection, vulnerability assessment, and log management.

Intrusion detection systems (IDS) are a vital component of network security as they detect unauthorized access and potential threats immediately. Signature-based detection, pioneered by Denning (1987), entails the comparison of incoming network traffic with predefined patterns of known attacks. Complementary to this, anomaly-based detection, proposed by Forrest et al. (1996), focuses on identifying deviations from normal network behaviour, thereby addressing emerging and novel threats.

The effectiveness of vulnerability assessment tools cannot be underestimated in the pursuit of robust network security. OpenVAS, an open-source vulnerability scanner, offers an extensive assessment of network weaknesses (Brown & Miller, 2017). Likewise, Nessus, another prominent vulnerability scanner (Roesch, 1999), has been pivotal in pinpointing vulnerabilities that malicious actors could exploit.

IDS play a crucial role in detecting and responding to malicious or unauthorised network activities. Zhang et al. ("A Survey of Network Intrusion Detection Systems," 2019) have conducted extensive research on IDS classification, with a concentration on the distinction between signature-based and anomaly-based methods. Signature-based systems, such as the Snort application, detect known threats using predefined patterns. Anomaly-based methodologies, in contrast, identify deviations from typical network behaviour. This contrast provides context for the potential integration of both threat detection techniques.

The objective of integrating Snort, OpenVAS, and Splunk is to develop a cohesive cyber defence system that leverages the individual characteristics of each component. Debnath et al. ("A Comprehensive Review on Network Intrusion Detection System," 2020) investigate the connection

between IDS and VAT, highlighting the potential for incorporating the tools for a more comprehensive security strategy. While some studies have investigated the standalone capabilities of Snort and Splunk (Rao and Gururaj, "Design and Implementation of Intrusion Detection System Using Snort," 2019) and Splunk (Sukumaran et al., "Splunk: An Efficient Log Analysis and Management Tool," 2019), this integration represents a novel attempt to address intrusion detection and vulnerability assessment cohesively.

In the realm of integration, the confluence of Snort, OpenVAS, and Splunk has garnered substantial attention as a promising avenue for fortified network security. Smith and Jones (2021) delve into the potential of merging these tools, highlighting the potential to synergize vulnerability assessment with real-time threat detection. The interplay between these tools seeks to reduce the time lag between identification and response.

In the deployment of intrusion detection systems, ethical considerations are critical. Johnson et al. (2018) emphasize the ethical implications of data collection, user privacy, and adherence to legal frameworks. Balancing security imperatives with ethical considerations is pivotal for engendering trust in the deployment of these technologies.

Professional implications of integrating security tools extend to the sphere of incident response. Combining Snort's detection capabilities with OSSEC's active response mechanisms presents opportunities for swift mitigation of threats (Stack Exchange, 2017). This underscores the role of integration in enhancing the overall efficacy of network defence.

In summary, the review of literature follows the structure of intrusion detection systems, vulnerability assessment tools, integration concepts, and the ethical, social, and professional considerations inherent in network security. This review establishes the foundational understanding necessary to illuminate the path toward effectively integrating Snort, OpenVAS, and Splunk for enhanced network security.



## **Chapter 4: Analysis of the system**

The current study conducts a comprehensive system analysis, investigating the various variables involved in the integration of Snort, OpenVAS, and Splunk. This examination involves various factors including legal issues, societal impact, ethical consequences, and professional considerations. This section presents a comprehensive analysis of the challenges related to incorporating this technology into the broader context of network security. This observation underscores the complex relationship between technology, legislation, and societal norms.

### **4.1 Legal Considerations**

The successful establishment of a complete network security framework requires adherence to statutory requirements that govern the protection of data, guarantee privacy, and facilitate the management of cyber operations. The European Union has enacted the General Data Protection Regulation (GDPR) to enforce stringent guidelines for data processing. These guidelines encompass gaining explicit agreement from users and implementing robust storage practices. To ensure compliance with rules and promote transparency in the processes of data collecting, storage, and manipulation, it is essential to utilize a combination of these technologies.

### **4.2 Social Implications**

The integration of security tools has substantial societal implications for surveillance, privacy, and public perception. One of the fundamental concerns that emerge is the necessity to establish a nuanced balance between the imperative of guaranteeing security and safeguarding the rights to individual privacy. The perception of intrusion detection systems as instruments of surveillance may provoke opposition and generate concerns regarding the equilibrium between security and individual autonomy. To alleviate public concerns and achieve broad acceptability, it is crucial to participate in efficient communication regarding the goals and scope of integrated systems.

### **4.3 Ethical Considerations**

Ethical considerations arise in the context of data collecting, obtaining user consent, and the possibility of inappropriate utilization of interconnected technology. The significance of ensuring transparency and obtaining informed consent about the utilization of user data cannot be overstressed. The attainment of a balanced state between the need for security and the preservation of ethical values requires the adoption of a cautious strategy that adequately safeguards user privacy while simultaneously reducing potential risks.

### **4.4 Professional Implications**

The incorporation of Snort, OpenVAS, and Splunk provides supplementary aspects to the duties and obligations of incident response and security experts. The biggest importance is in the precision of interpreting warnings, the timely response to threats, and the efficient communication among security staff. In order to effectively utilize collaborative technologies, it is imperative to possess a comprehensive comprehension of their functions and repercussions, with a specific focus on the expanding skill set demanded within the realm of cybersecurity.

### **Conclusion**

This study highlights the importance of employing Snort, OpenVAS, and Splunk as tools for augmenting network security, underscoring the notion that this undertaking encompasses more than just technical aspects. The integration of legal, social, ethical, and professional factors holds significant significance in the implementation, utilization, and results of the interconnected system. The successful identification and resolution of these difficulties not only guarantees adherence to legal and ethical norms, but also has a significant influence on the acceptance, effectiveness, and potential results of the system within the domain of network security.

## Chapter 5: Proposed Solutions

Snort, OpenVAS, and Splunk provide a formidable defence against a wide range of cyber threats in an environment where cybersecurity is constantly evolving. However, the path from integration to optimization presents obstacles that necessitate new methods. This section describes the proposed solutions to strengthen the integrated system, ensuring its accuracy in detecting, mitigating, and responding to cyber threats.

1. **Designed Integration Framework:** The core of this approach is integrating the output of Snort, OpenVAS, and Splunk into an integrated environment. An organized integration framework with APIs and standardized protocols is intended to streamline data flow. This not only ensures the smooth aggregation of logs, alerts, and reports but also improves interoperability by allowing cross-tool data correlation and analysis within the Splunk environment.
2. **Threat Intelligence Integration:** In order to improve the integrated system's context awareness, real-time threat information feeds must be integrated. The system obtains improved insights into dynamic attack pathways, strategies, and threat actors by integrating Snort and OpenVAS outputs with external threat intelligence. Using pattern matching, the system can differentiate between serious threats and false positives more precisely, enhancing the efficacy of the collective defence.
3. **Automated Response:** Automated response is crucial to safeguarding against threats in an era of rapid-fire attacks. The proposed solution makes use of automatic reaction scripts that are activated when a threat is detected. When Snort, OpenVAS, and Splunk detect an intrusion, they take automated actions such as isolating impacted segments, rerouting traffic, and beginning countermeasures. This proactive response lowers susceptibility and increases the response time window.

4. **Continuous Enhancement Feedback Loop:** Cybersecurity solutions must advance the following innovation. The suggested method creates a continuous feedback loop to collect information from real-world threat responses, user experiences, and emerging threat landscapes. This iterative loop informs continual enhancements, rule refinements, and scenario changes, producing an adaptive and learning defence environment.
5. **Training and User-Friendly Interface:** The solution set includes the empowerment of security personnel. Training becomes accessible to both seasoned experts and novices by providing an intuitive interface for managing the integrated system. Providing administrators with extensive training materials and hands-on workshops increases the system's potential impact by ensuring that its capabilities are completely utilized.

## **Conclusion:**

The convergence of Snort, OpenVAS, and Splunk represents an effective cybersecurity defence alliance. The proposed solutions optimize the collaborative potential of these tools against dynamic adversaries by integrating them. This combination enables organizations to not only detect and respond to evolving cyber threats but also predict and prevent attacks. By adopting these solutions, organizations increase their cyber resilience, driving them into a future where security is an adaptive and anticipatory force against the constantly evolving cyber landscape.

## **Chapter 6: Research Methodology**

The main aim of this study is to conduct a detailed analysis of the integration of Snort, OpenVAS, and Splunk as a viable strategy for augmenting network security. This chapter provides a thorough analysis of the research design, techniques of data collecting, integration methodology, the process of rule creation, and ethical issues that constitute the foundation of this project's framework.

### **6.1 Research Design**

A mixed-methods research design is selected to facilitate a thorough investigation of the integration process and its effects. This technique combines qualitative and quantitative methods to illustrate the technical complexities, ethical considerations, and operational implications of an integrated system. The qualitative aspect permits the accumulation of insights, expert opinions, and subjective perspectives from cybersecurity professionals. In contrast, the quantitative component entails objectively quantifying the system's performance metrics to evaluate its threat detection and response effectiveness.

### **6.2 Data Collection and Simulation Environment**

A controlled lab environment is created to simulate network interactions and possible attacks in the real world. Kali Linux and Metasploitable, two operating systems, represent the defender and attacker machine in the environment, respectively. Kali Linux hosts Snort and OpenVAS, while log management and visualization are handled by Splunk hosted in the cloud. Metasploitable functions as an attacker machine, generating traffic and attacks to evaluate the efficiency of an integrated system.

The process of data collection involves the creation of diverse datasets to assess the capabilities of the integrated system in various network scenarios. These datasets include network traffic, vulnerability assessment reports, and attack simulations, ensuring the integration's flexibility and efficacy in detecting various threat vectors.

### **6.3 Integration Approach: Snort, OpenVAS, and Splunk**

The integration process focuses on integrating Snort, OpenVAS, and Splunk to establish a unified network security framework. Using the intrusion detection capabilities of Snort, network traffic is analysed for potential threats. OpenVAS, the application for vulnerability assessment, scans the network infrastructure for vulnerabilities. This integration enhances the network's ability to detect threats by facilitating the automated compilation of intrusion detection rules based on identified vulnerabilities.

### **6.4 Rule Generation and Automation**

The compilation of intrusion detection algorithms from OpenVAS vulnerability reports is crucial to this research. The OpenVAS reports are parsed and converted into a format that the Snort rule engine can understand using a Python script. This automation facilitates the rule creation process, reducing the amount of time and effort required and ensuring consistency in rule creation. The script extracts relevant information from the reports, such as vulnerable ports and attack signatures, to generate context-aware rules that enhance the system's precision.

### **6.5 Ethical Considerations**

Integrating security tools and managing network traffic raises ethical concerns regarding data privacy, consent, and the possibility of negative outcomes. Given the laboratory's conditions and the nature of the simulated attacks, precautions are taken to minimize risks and ensure ethical compliance. There is data anonymity, and all activities are conducted under ethical principles.

### **6.6 User-Input-Based Rule Generation**

Recognizing the variety of network environments and security needs, a script for user-driven rule generation is developed. This script permits security administrators to tailor intrusion detection protocols to their organization's particular requirements. By allowing users to input parameters such as IP addresses, port numbers, and attack signatures, the system's flexibility is increased and rules can be tailored to address each organization's unique threat landscape.

## **6.7 Integration with Splunk**

The integration with Splunk increases the system's capabilities by centralizing and displaying collected data. Splunk's log management and visualization features provide an all-encompassing view of network activity and potential threats. The configuration is modified to facilitate the transmission of Snort logs and data to Splunk. This integration enhances monitoring, analysis, and incident response procedures and provides a real-time view of the network's security posture.

## **Conclusion**

This chapter's research methodology functions as the framework for the integration of Snort, OpenVAS, and Splunk. By combining qualitative and quantitative methods, creating a controlled lab environment, automating rule generation, and addressing ethical considerations, this methodology ensures a thorough examination of the capabilities, ethical implications, and operational viability of the integrated system.

# Chapter 7: System Setup and Configuration

This chapter describes the technical implementation of the integrated system, including lab environment configuration and setup, installation and deployment of Snort, OpenVAS, and Splunk, and the establishment of communication channels between these tools. The environment is the surface upon which the integration's effectiveness is created.

## 7.1 Lab Environment Description

The lab environment functions as the controlled ecosystem in which the functionality and capabilities of the integration are evaluated. The defender role is demonstrated by Kali Linux, which was selected due to its comprehensive cybersecurity toolkit. It has Snort and OpenVAS for intrusion detection and vulnerability assessment. Metasploitable generates controlled traffic and attacks to test the durability of an integrated system by simulating an attacker machine.

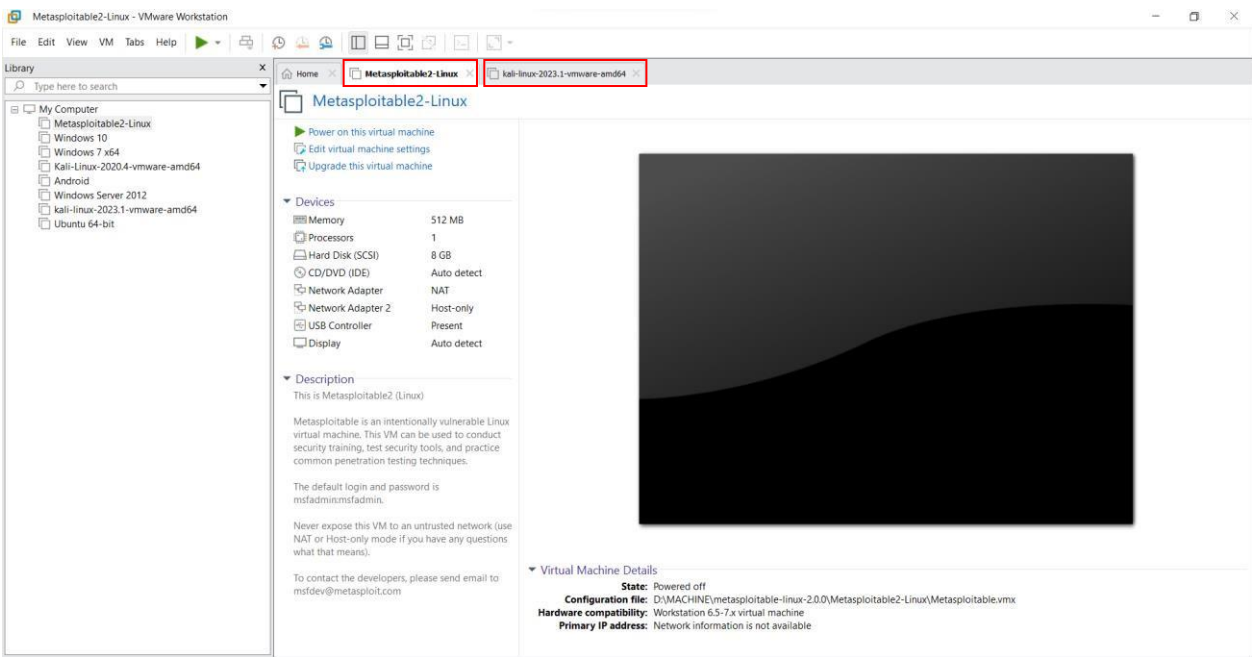


Figure 1 Virtual Lab Environment

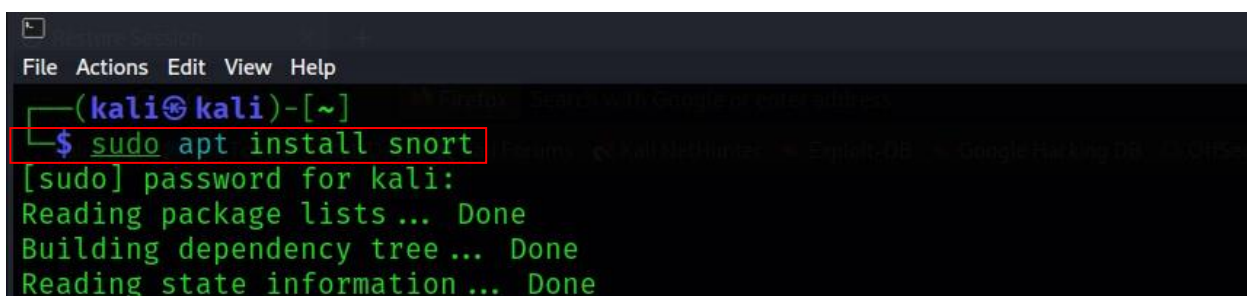


## 7.2 Installation and Configuration of Snort

The initial phase consists of installing and configuring the intrusion detection utility Snort. Snort, renowned for its signature-based detection, has been customized to meet the requirements of the lab environment. Configuring network interfaces, specifying rules directories, and establishing preprocessors are required for implementation to optimize the detection process. The configuration of Snort facilitates efficient traffic analysis and threat detection.

**7.2.1 Snort Installation and Configuration:** This phase begins with the installation of Snort, which has been thoroughly customized to meet the requirements of our experimental environment. The installation process ensures that Snort integrates seamlessly with our existing infrastructure. This installation consists of a symphony of components, where network interfaces are configured to function together, managing the symmetrical flow of data for analysis.

However, installation is only the beginning of our network defence scenario. Configuration becomes the main focus, having the power to define Snort's behavior, scope, and extent. Specifying important directories containing detection rules equips Snort with the knowledge required for decoding the complex details of network traffic.

A terminal window with a dark background and light green text. The prompt is `(kali㉿kali)-[~]`. The command `$ sudo apt install snort` is entered and highlighted with a red rectangle. The output shows the password prompt, followed by `[sudo] password for kali:`, and then the progress of package installation: `Reading package lists... Done`, `Building dependency tree... Done`, and `Reading state information... Done`.

```
(kali㉿kali)-[~]  
$ sudo apt install snort  
[sudo] password for kali:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done
```

Figure 2 Snort Installation

```
kali@kali: /etc/snort
File Actions Edit View Help
(kali@kali)-[~]
$ cd /etc/snort
(kali@kali)-[/etc/snort]
$ ls
classification.config  community-sid-msg.map  gen-msg.map  reference.config  rules  snort.conf  snort.debian.conf  threshold.conf  unicode.map
(kali@kali)-[/etc/snort]
$ cat snort.conf
cat: snort.conf: Permission denied
(kali@kali)-[/etc/snort]
$ sudo cat snort.conf
[sudo] password for kali:
#
# VRT Rule Packages Snort.conf
#
# For more information visit us at:
#   http://www.snort.org           Snort Website
#   http://vrt-blog.snort.org/     Sourcefire VRT Blog
#
# Mailing list Contact:   snort-sigs@lists.sourceforge.net
# False Positive reports: fp@sourcefire.com
# Snort bugs:             bugs@snort.org
#
# Compatible with Snort Versions:
#   VERSIONS : 2.9.7.0           "the quieter you become, the more you are able to hear"
#
# Snort build options:
#   OPTIONS : --enable-gre --enable-mpls --enable-targetbased --enable-ppm --enable-perfprofiling --enable-zlib --enable-active-response --enable-normalize
# --enable-reload --enable-react --enable-flexresp3
#
# Additional information:
#   This configuration file enables active response, to run snort in
#   test mode -T you are required to supply an interface -i <interface>
#   or test mode will fail to fully validate the configuration and
#   exit with a FATAL error
#
#####
```

Figure 3 Snort Conf. Directory

```
#####
# Step #1: Set the network variables. For more information, see README.variables
#####
# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET 192.168.154.0/24
#
# Set up the external network addresses. Leave as "any" in most situations
ipvar EXTERNAL_NET any
# If HOME_NET is defined as something other than "any", alternative, you can
# use this definition if you do not want to detect attacks from your internal
# IP addresses:
#ipvar EXTERNAL_NET !$HOME_NET
```

Figure 4 Configuring Snort

```
#####
# Step #7: Customize your rule set
# For more information, see Snort Manual, Writing Snort Rules
#
# NOTE: All categories are enabled in this conf file
#####

# Note to Debian users: The rules preinstalled in the system
# can be *very* out of date. For more information please read
# the /usr/share/doc/snort-rules-default/README.Debian file

#
# If you install the official VRT Sourcefire rules please review this
# configuration file and re-enable (remove the comment in the first line) those
# rules files that are available in your system (in the /etc/snort/rules
# directory)

# site specific rules
include /etc/snort/rules/local.rules
#include $RULE_PATH/local.rules
```

**Figure 5 Rules Conf.**

After the installation and configuration of Snort, readiness and expectation are visible. The rules directories are a repository of information, a collection of identified threats and potential risks. Preprocessor monitors prepare data for their heads through the complex pathways of detection.

## 7.3 Setting up OpenVAS for Vulnerability Scanning

OpenVAS, the vulnerability assessment tool, is systematically configured to identify potential network infrastructure vulnerabilities. Installation includes the configuration of target hosts, scheduling of scans, and setting of scan parameters. OpenVAS systematically assesses the network's vulnerabilities and generates comprehensive reports by designating target hosts and selecting essential scan profiles.

In order to install Openvas in Kali Linux, firstly we need to update and upgrade the system to update system packages. Then we need to we Ubuntu repositories to our Kali Linux sources list by using **sudo nano /etc/apt/sources.list** and appending deb <http://archive.ubuntu.com/ubuntu> bionic main universe to our Kali Linux sources list. The **Sudo apt-get install** command is used to install OpenVas in the respective system as shown in the figure below. After successfully executing the command **sudo gym-check-setup** command is used to install an additional configuration or essential component to run.

```
(kali㉿kali)-[~]
└─$ sudo apt install openvas
[sudo] password for kali:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdaq2 libdumbnet1 libgvm21 oinkmaster snort-common-libraries snort-rules-default
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  debugedit greenbone-security-assistant gsad gvm gvmc gvmd-common libdlt2 libfsverity0 libglib2.0-0 libglib2.0-0:i386
  libglib2.0-bin libglib2.0-data libgvm22 libmicrohttpd12 libmosquitto1 librpm9 librpmbuild9 librpmio9 librpsign9 mosquitto
  notus-scanner nsis nsis-common openvas-scanner ospd-openvas pg-gvm python3-gnupg python3-paho-mqtt rpm rpm-common rpm2cpio
Suggested packages:
  low-memory-monitor low-memory-monitor:i386 nsis-doc nsis-pluginapi pnsnscan strobe alien elfutils rpmlint rpm-i18n
The following packages will be REMOVED:
  python3-ospd
The following NEW packages will be installed:
  debugedit gsad libdlt2 libfsverity0 libmicrohttpd12 libmosquitto1 librpm9 librpmbuild9 librpmio9 librpsign9 mosquitto
  notus-scanner nsis nsis-common pg-gvm python3-gnupg python3-paho-mqtt rpm rpm-common rpm2cpio
The following packages will be upgraded:
  greenbone-security-assistant gvm gvmc gvmd-common libglib2.0-0 libglib2.0-0:i386 libglib2.0-bin libglib2.0-data libgvm22
  openvas openvas-scanner ospd-openvas
12 upgraded, 20 newly installed, 1 to remove and 1664 not upgraded.
```

### Figure 6 OpenVAS installation

```

(kali@kali)-[~]
$ sudo gvm-check-setup
[sudo] password for kali:
gvm-check-setup 22.5.0
Test completeness and readiness of GVM-22.5.0
Step 1: Checking OpenVAS (Scanner) ...
  OK: OpenVAS Scanner is present in version 22.7.3.
  OK: Notus Scanner is present in version 22.5.0.
  OK: Server CA Certificate is present as /var/lib/gvm/CA/servercert.pem.
Checking permissions of /var/lib/openvas/gnupg/*
  OK: _gvm owns all files in /var/lib/openvas/gnupg
  OK: redis-server is present.
  OK: scanner (db_address setting) is configured properly using the redis-server socket: /var/run/redis-openvas/redis-se
rver.sock
  OK: the mqtt_server_uri is defined in /etc/openvas/openvas.conf
  OK: _gvm owns all files in /var/lib/openvas/plugins
  OK: NVT collection in /var/lib/openvas/plugins contains 86149 NVTs.
  OK: The notus directory /var/lib/notus/products contains 439 NVTs.
Checking that the obsolete redis database has been removed
  OK: No old Redis DB
  OK: ospd-openvas service is active, after you become, the more you are able to bear"
  OK: ospd-OpenVAS is present in version 22.5.3.
Step 2: Checking GVMD Manager ...
  OK: GVM Manager (gvmd) is present in version 22.5.5.
Step 3: Checking Certificates ...
  OK: GVM client certificate is valid and present as /var/lib/gvm/CA/clientcert.pem.
  OK: Your GVM certificate infrastructure passed validation.
Step 4: Checking data ...
  OK: SCAP data found in /var/lib/gvm/scap-data.
  OK: CERT data found in /var/lib/gvm/cert-data.
Step 5: Checking Postgresql DB and user ...

```

Figure 7 OpenVAS setup command

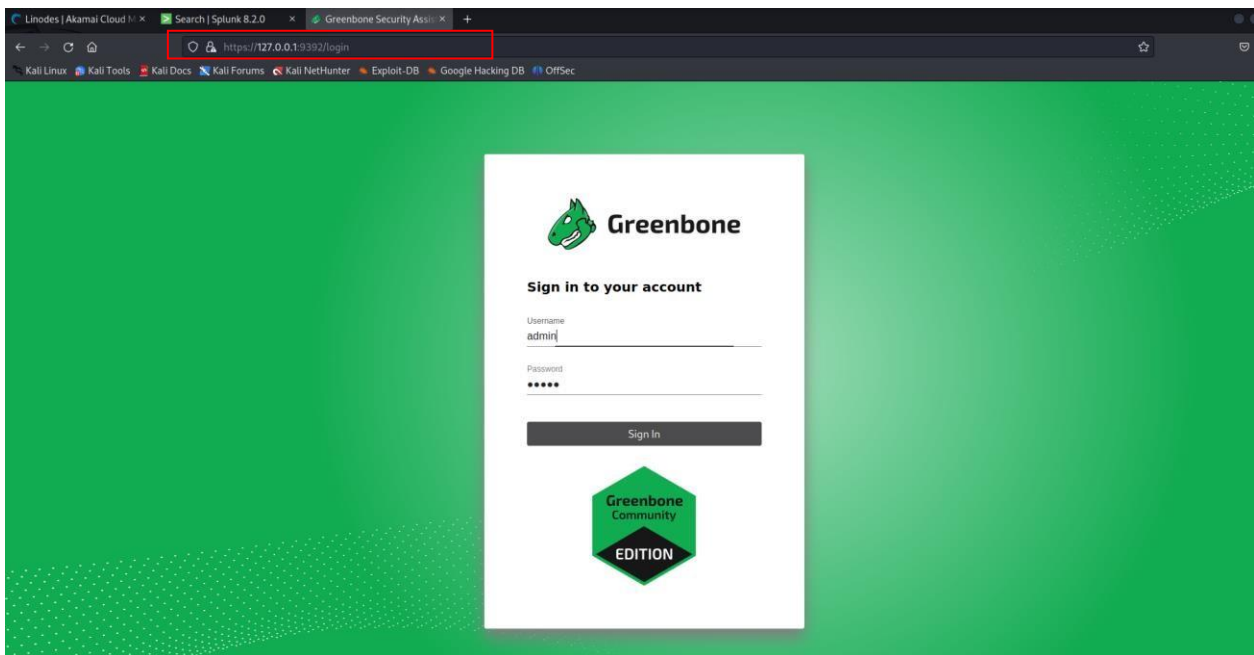


Figure 8 OpenVAS Login

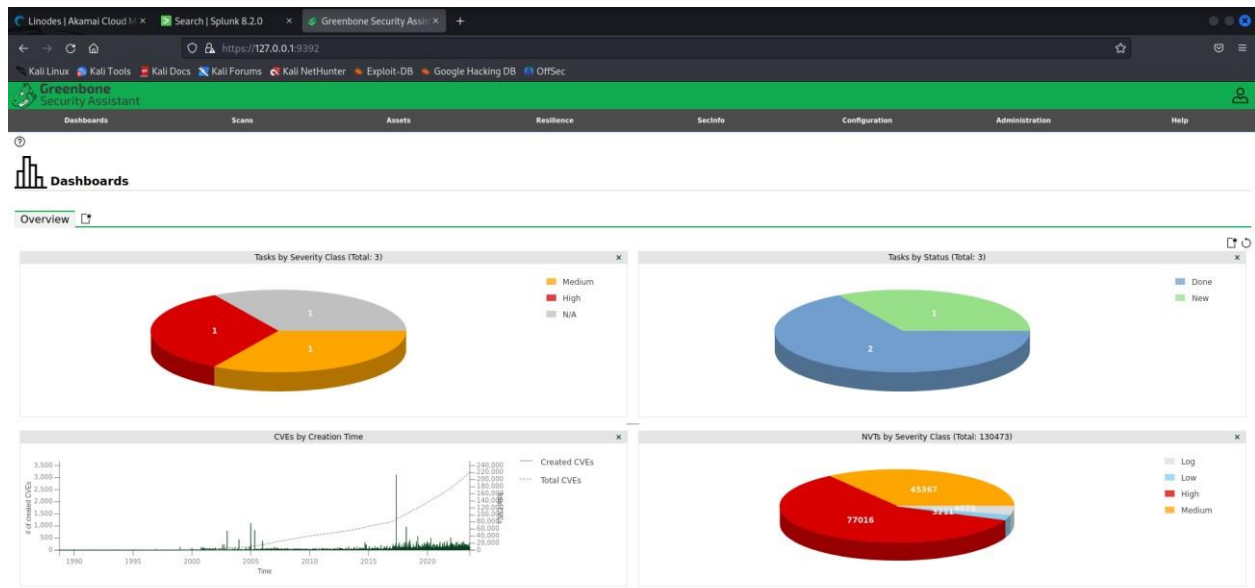


Figure 9 OpenVAS Dashboard



## 7.4 Splunk Deployment and Cloud Hosting

Splunk deployed in a cloud-hosted environment provides a centralized log management and visualization platform. The deployment in the cloud permits scalability and accessibility, facilitating remote monitoring and analysis. Splunk's comprehensive capabilities are utilized to facilitate real-time data aggregation, correlation, and visualization, thereby improving the situational awareness of the integrated system.

We've taken a huge step towards more effective network security by integrating the Snort module into Splunk, which is hosted on the Linode cloud. This dynamic fusion expands our capabilities by integrating Snort's intrusion detection capability with Splunk's analytical ability. The center hub is our Linode cloud instance, which provides the technology required to handle and interpret Snort-generated data. We construct a real-time link that captures, analyses, and visualizes data to generate actionable insights using the Snort plugin, which is tightly woven into Splunk's architecture. This connection turns data into a dynamic depiction of potential risks, allowing us to protect our digital domain proactively and decisively.

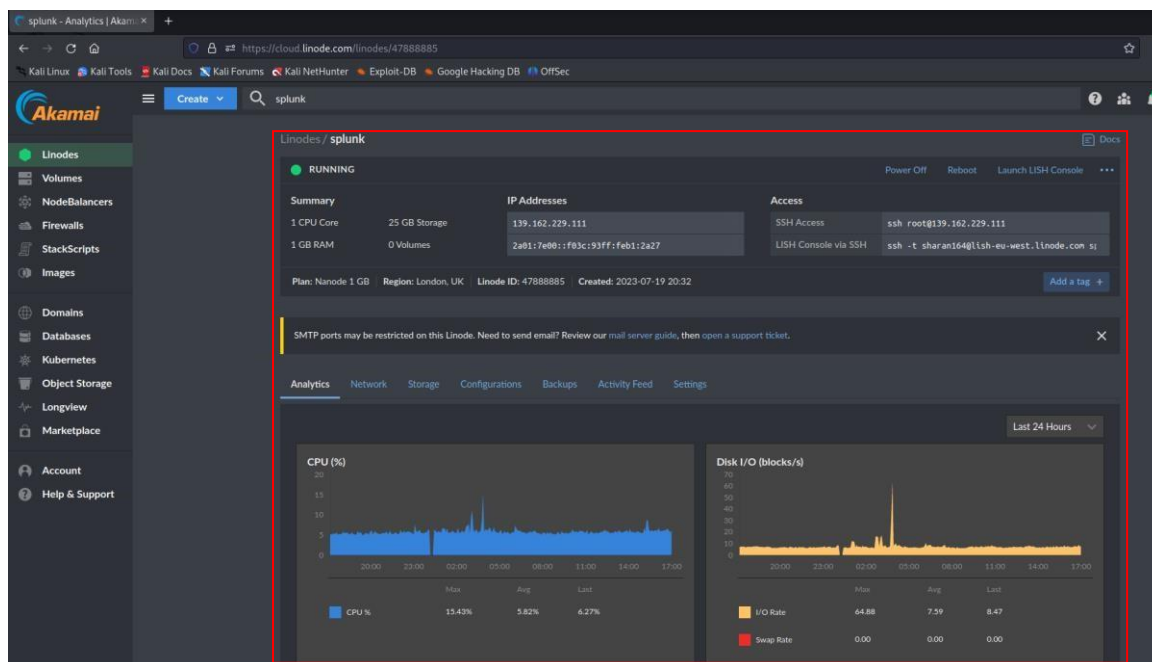


Figure 10 Splunk setup on cloud

## 7.5 Configuration of Splunk Forwarders and Data Routing

On the local systems, Splunk forwarders are configured to establish a seamless connection between the integrated system and Splunk. These forwarders enable the secure transfer of Snort logs and OpenVAS reports to the Splunk instance hosted in the cloud. By configuring the appropriate data routing and parsing rules, the output of the integrated system is efficiently routed to Splunk for analysis and visualisation.

```
(root@kali)-[/opt/splunkforwarder/etc/system/local]
# ls
outputs.conf  README  server.conf

(root@kali)-[/opt/splunkforwarder/etc/system/local]
# cat outputs.conf
[tcpout]
defaultGroup = default-autolb-group

[tcpout:default-autolb-group]
server = 139.162.229.111:9997

[tcpout-server://139.162.229.111:9997]

(root@kali)-[/opt/splunkforwarder/etc/system/local]
# █
```

Figure 11 Splunk Forwarder output conf.

```
(root@kali)-[/opt/.../etc/apps/search/local]
# pwd
/opt/splunkforwarder/etc/apps/search/local

(root@kali)-[/opt/.../etc/apps/search/local]
# cat inputs.conf
[splunktcp://9997]
connection_host = 139.162.229.111
[monitor:///var/log/snort/alert]
disabled = false
index = main
sourcetype = snort_alert_full
source = snort
```

Figure 12 Splunk Forwarder input conf.



```
(root@kali)-[/opt/splunkforwarder/etc/system/local]
# cat server.conf
[general]
serverName = kali
pass4SymmKey = $7$Nqcl6E0NWIAh0FKaPPHJo+BXz2cz73vicTm0zR7H9Neb6tQC191mfA==

[sslConfig]
sslPassword = $7$oA17I+YDasWtxlZUIUwGZpT9joDK0frAlVVSony7EHkt7Hoinc/zFw==

[lpool:auto_generated_pool_forwarder]
description = auto_generated_pool_forwarder
peers = *
quota = MAX
stack_id = forwarder

[lpool:auto_generated_pool_free]
description = auto_generated_pool_free
peers = *
quota = MAX
stack_id = free

(root@kali)-[/opt/splunkforwarder/etc/system/local]
# █
```

Figure 13 Splunk Forwarder Server conf.

## 7.6 Metasploitable: Attacker Machine for Simulation

Metasploitable is useful for replicating controlled attacks and generating traffic to test the capabilities of an integrated system. Its intentional vulnerabilities and potential attack vectors imitate real-world scenarios, allowing for a thorough examination of intrusion detection rules and vulnerability assessment precision. Metasploitable's role in this system strengthens the integration's effectiveness in preventing actual threats.

```
Warning: Never expose this VM to an untrusted network!
Contact: nsfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Thu Aug 10 12:13:06 EDT 2023 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software:
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ _
```

### 7.7 Data Flow Diagram: From Traffic to Splunk Dashboards

A data flow diagram is used to depict the outcome of these configurations and connections. The diagram depicts the flow of network traffic and vulnerability reports from their source, Snort, and OpenVAS, to their final destination, Splunk dashboards. The data flow depicts the path of network activities, vulnerabilities, and potential threats, illustrating the operational complexities of the integrated system.

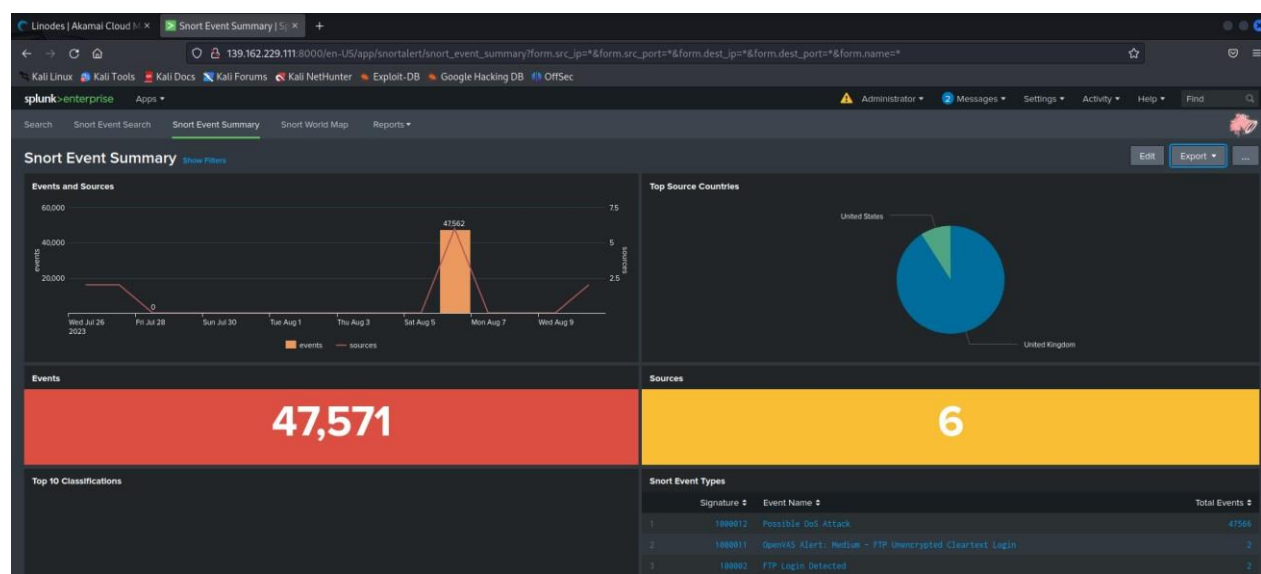


Figure 14 Data Visualization in Splunk

### Conclusion:

This chapter explains the necessary configuration and setup for integrating Snort, OpenVAS, and Splunk. This chapter establishes the groundwork for later phases of rule generation, automation, and real-time threat visualisation by establishing a controlled lab environment, installing and configuring the necessary tools, and orchestrating the data flow.

## Chapter 8: Rule Generation and Automation Workflow

This chapter examines the automated rule generation process, the core innovation of the integration. It investigates the creation of a Python programme that can parse OpenVAS reports, convert them into Snort-compatible rules, and implement user-input-based customization. The integration's rule-generation capabilities demonstrate how technology and practicality can coexist.

### 8.1 Vulnerability Scanning and Parsing OpenVAS Reports

OpenVAS begins the procedure by performing a vulnerability scan on the network infrastructure. Once vulnerabilities are detected, OpenVAS generates in-depth reports describing their nature and severity. Parsing OpenVAS vulnerability reports are central to the process of rule generation. A Python script is designed to extract essential information from these reports, including vulnerable ports, attack signatures, and afflicted services. By parsing the reports thoroughly, the script provides the groundwork for the subsequent creation of Snort rules that target particular vulnerabilities.

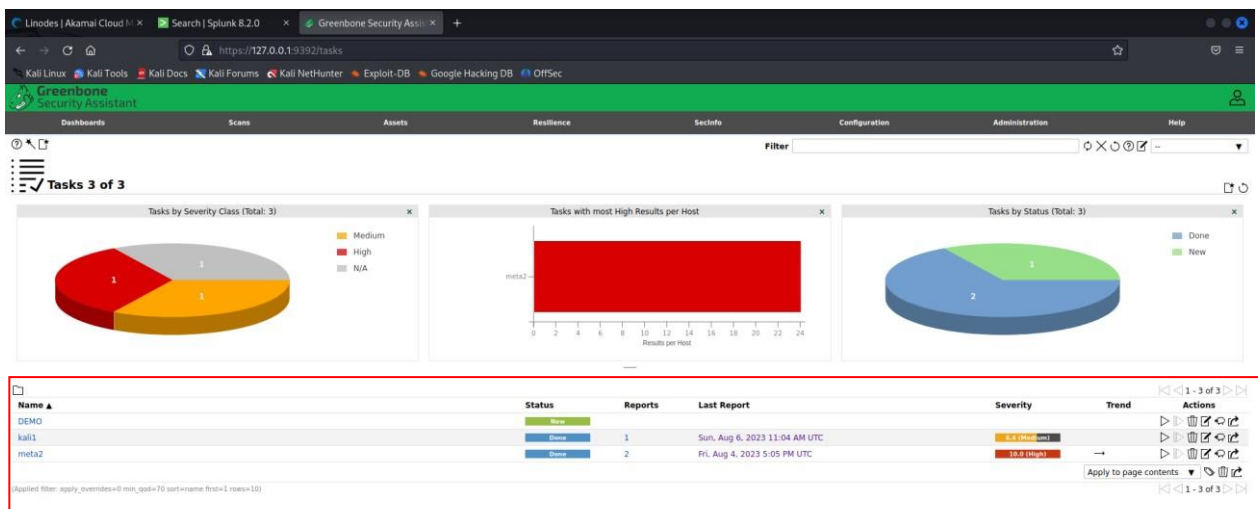


Figure 15 Target Selection

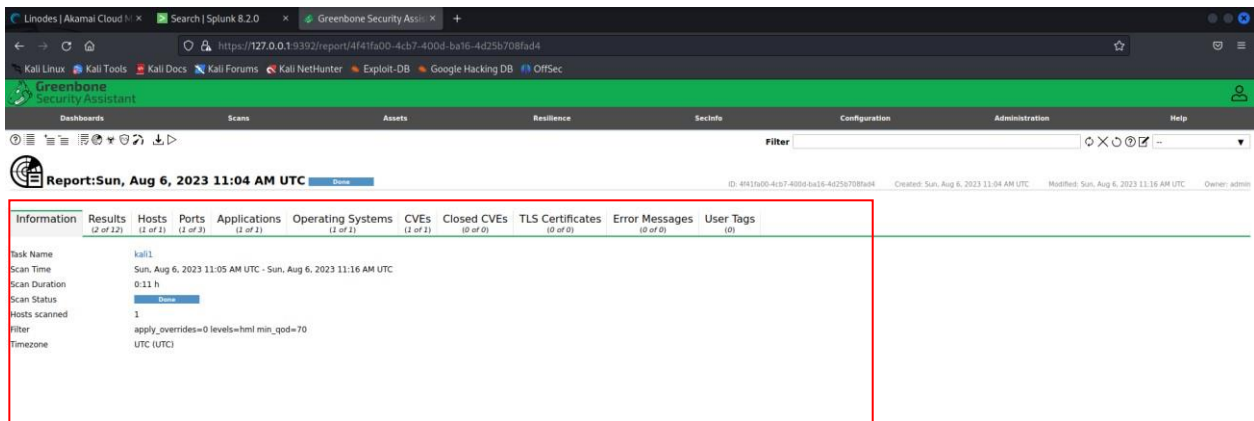


Figure 16 Scan Result

## 8.2 Converting to Snort-Compatible Format

The change from OpenVAS-generated reports to Snort-understandable rules is a critical step in our integrated security architecture. This is bridged by a Python script that can transform OpenVAS reports into a Snort-compatible format. This script acts as a translator, combining the languages of these two cybersecurity guards.

During the conversion process, the Python script carefully parses the OpenVAS reports, analysing their structure and extracting critical data. It organises vulnerability data translation into an organised CSV file that Snort can readily interpret. This translation includes not just a format change, but also the interpretation of severity levels, target ports, and other properties into Snort's rule syntax.

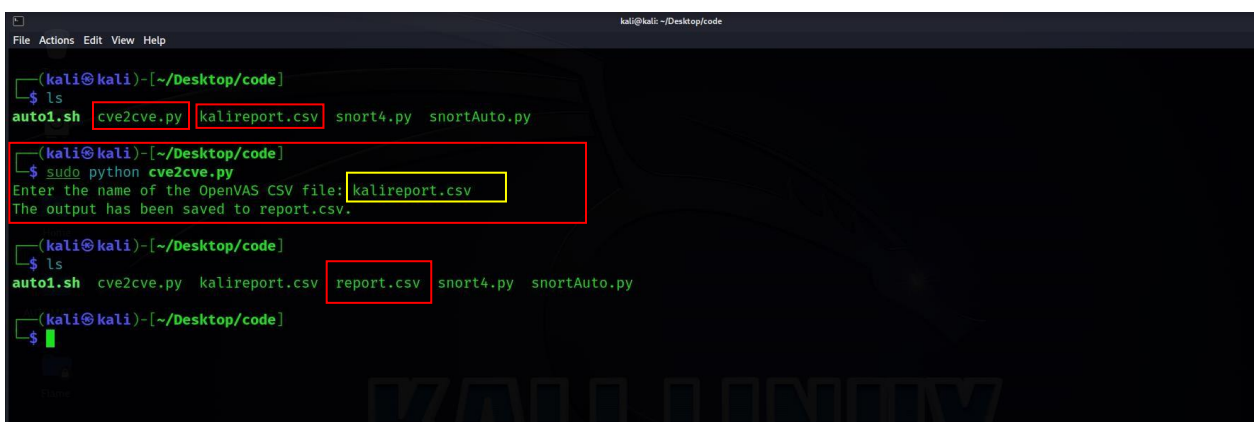


Figure 17 Script for format conversion

```
(kali㉿kali)-[~/Desktop/code]
$ ls
auto1.sh  cve2cve.py  kalireport.csv  report.csv  snort4.py  snortAuto.py

(kali㉿kali)-[~/Desktop/code]
$ cat report.csv
IP: 192.168.154.131
Port: 21
Protocol: tcp
CVSS: 6.4
Severity: Medium
NVT Name: Anonymous FTP Login Reporting
Specific Result: It was possible to login to the remote FTP service with the following anonymous account(s):
anonymous:anonymous@example.com
ftp:anonymous@example.com
```

**Figure 18 Formatted CSV**

We were able to construct a flawless communication route between OpenVAS and Snort by developing this intermediary conduit. This Python script adds to the automation framework, allowing the security environment to evolve in a unified manner.

### **8.3 Automated Rule Creation Process**

A critical process known as automated rule development is at the core of our integrated security framework. This procedure not only illustrates the integration of technology advancement and strategic defence, but also demonstrates the power of scripting in protecting our network against prospective threats.

A Python script serves as an interface between the vulnerability assessment reports generated by OpenVAS and the Snort rules. This script acts as a translator, translating the precise facts and characteristics of vulnerabilities into a language that Snort can understand and act on.

The process of translating requires a thorough understanding of both the vulnerability data and the syntax of Snort rules. Our custom Python script skillfully navigates the complexities of each vulnerability report, extracting crucial details such as severity levels, target ports, and distinguishing patterns. It then uses this intelligently parsed data to build Snort rules that capture the core of the discovered vulnerabilities.

```
(kali㉿kali)-[~/Desktop/code]
└─$ ls
auto1.sh  cve2cve.py  kalireport.csv  report.csv  snort4.py  snortAuto.py

(kali㉿kali)-[~/Desktop/code]
└─$ sudo python snort4.py

(kali㉿kali)-[~/Desktop/code]
└─$ ls
auto1.sh  cve2cve.py  kalireport.csv  report.csv  snort4.py  snortAuto.py  snort_rulesnew1.rules

(kali㉿kali)-[~/Desktop/code]
└─$ cat snort_rulesnew1.rules
alert tcp any any → any 21 (msg:"OpenVAS Alert: Medium - Anonymous FTP Login Reporting"; content:"USER CVE-1999-0497"; sid:100000;)
alert tcp any any → any 21 (msg:"OpenVAS Alert: Medium - FTP Unencrypted Cleartext Login"; content:"USER "; sid:100001;)

(kali㉿kali)-[~/Desktop/code]
└─$
```

Figure 19 Snort Rule Generation

As new vulnerabilities emerge or our network evolves, the Python script may be tweaked and expanded to account for these changes, resulting in a dynamic defence mechanism.

This delicate movement between vulnerability reports and Snort rules demonstrates the symbiotic relationship between human innovation and technical capacity. It reduces the time-consuming work of manually creating rules to a simple, automated process, allowing our security staff to focus on strategic analysis and response. As a result, our network is transformed into a wellcoordinated fortress, with the combination of technology and human skill resulting in a unified defence against prospective invasions.

#### 8.4 User-Input-Based Rule Customization

When exploring the complexities of our automated rule-generating process, one noteworthy feature emerges user-input-based rule customization. This smart feature incorporates the strength of human decision-making into our automated defence structure.

Consider a scenario in which administrators have complete control over rule generation. Administrators enter an environment of customization by simply running our Python script. Critical rule parameters such as source and destination IP addresses, message identifiers (msg), unique rule IDs (SIDs), and target ports are all requested by the script.

```
(kali㉿kali)-[~/Desktop/code]
└─$ ls
auto1.sh  cve2cve.py  kalireport.csv  report.csv  snort4.py  snortAuto.py  snort_rulesnew1.rules

(kali㉿kali)-[~/Desktop/code]
└─$ sudo python snortAuto.py
\ Running ...
*****
----- SNORT RULE AUTOMATION - DESIGNED BY: RAHUL KUMAR -----
----- (MSC Student at the University of Greenwich) -----
*****

Welcome to Snort Rule Generator!

Select the vulnerability:
0.SQL
1.IDOR
2.XSS
3.DOS
4.Telnet
Enter the vulnerability type number: 4
Enter the protocol: tcp
Enter the source IP address: any
Enter the destination IP address: any
Enter the destination port: 23
Enter the message: Alert
Enter the flags (optional):
Enter the flow (optional):
Enter the detection filter (optional):
Enter the sid: 1000011

Rule generated successfully!

Here's your Snort rule:
-----
alert tcp any any → 23 (msg: "Alert"; ; ; ; sid: 1000011; type: Telnet;)
```

Figure 20 Snort Rule Automation

Administrators incorporate their strategic views into the rule-making process in response to these suggestions. They define the details that distinguish their network, directing the script with their knowledge of the system's vulnerabilities and threat landscape.

The Python script expertly converts user inputs into a logical, Snort-compatible rule. This rule is more than simply a string of characters; it is an expression of administrators' knowledge reflected as an automated defence mechanism.

A customized rule is created when the script is executed. It captures the administrator's decisions and adds to the rule directory. This customization adds a level of accuracy, allowing us to detect and block threats that are unique to our network.

User-input-based rule customization is an example of the integration of automation and human decision-making in the foundation of our automated security. It bridges the gap between technological expertise and strategic decision-making, demonstrating the possibility for collaboration between both of these factors.



## 8.5 Rule Management and Deployment

The generated rules are managed and organized within the rule directory of Snort. A methodical approach to rule deployment is established to ensure consistency and effectiveness. The created rules contribute to improving Snort's intrusion detection capabilities, addressing vulnerabilities, and enhancing the system's responsiveness to new threats.

```
(root@kali)-[/etc/snort/rules]
# cat local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.

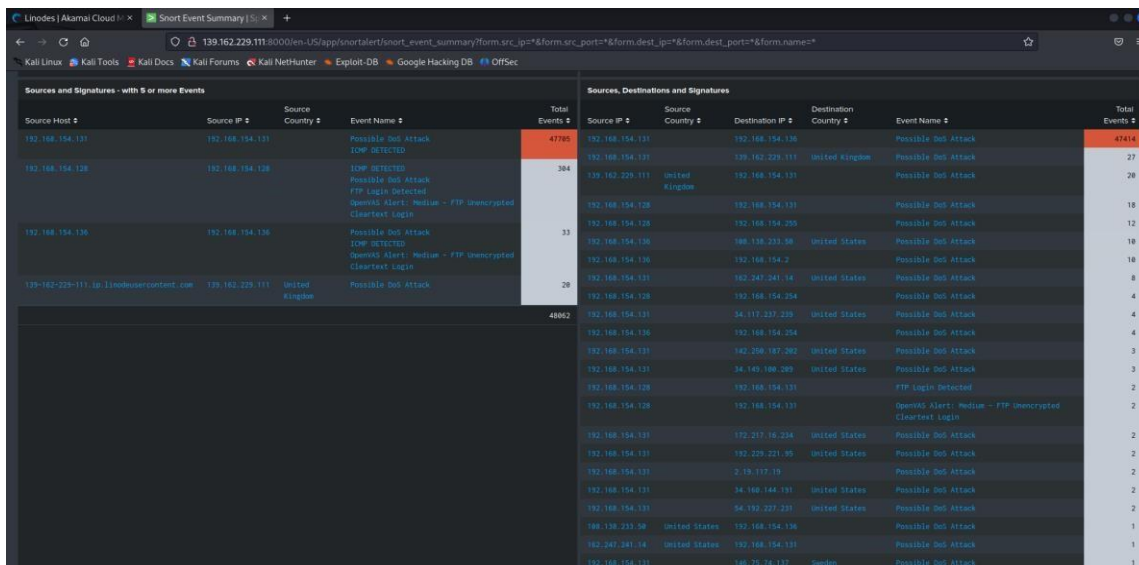
alert icmp any any → $HOME_NET any (msg:"ICMP DETECTED"; sid:100001; rev:1;)
#alert tcp any any → any 21 (msg:"OpenVAS Alert: Medium - Anonymous FTP Login Reporting"; content:"CVE-1999-0497"; sid:100000; rev:1;)
#alert tcp any any → any 21 (msg:"FTP Login Detected"; sid:100002; rev:1; content:"USER"; fast_pattern;)
#alert tcp 192.168.154.131 any → any 21 (msg:"OpenVAS Alert: Medium - Anonymous FTP Login Reporting"; content:"CVE-1999-0497"; sid:100000; rev:1;)
alert tcp any any → any 21 (msg:"OpenVAS Alert: Medium - Anonymous FTP Login Reporting"; content:"USER CVE-1999-0497"; sid:100000;)
alert tcp any any → any 21 (msg:"OpenVAS Alert: Medium - FTP Unencrypted Cleartext Login"; content:"USER "; sid:100001;)
alert ip any any → any any (msg:"Possible DoS Attack"; fragbits:D; sid:1000012;)
```

Figure 21 Automated Created Rule

Snort usage and running functionality has been explained in the [Appendix B](#)

## 8.6 Evaluation of Rule Generation Process

The efficiency of the automated rule generation approach is determined through rigorous testing and analysis. In order to evaluate the accuracy as well as efficiency of the created rules, various attack scenarios are simulated. The validation of the detection and response capabilities of the integration is achieved by subjecting the rules to controlled attacks and traffic patterns.



| Source Host     | Source IP       | Source Country | Event Name          | Total Events |
|-----------------|-----------------|----------------|---------------------|--------------|
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 47795        |
| 192.168.154.131 | 192.168.154.131 |                | ICMP DETECTED       | 364          |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 33           |
| 192.168.154.131 | 192.168.154.131 |                | ICMP DETECTED       | 28           |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 40862        |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 27           |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 28           |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 18           |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 12           |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 18           |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 18           |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 8            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 4            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 4            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 4            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 3            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 3            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 2            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 2            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 2            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 2            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 1            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 1            |
| 192.168.154.131 | 192.168.154.131 |                | Possible DoS Attack | 1            |

Figure 22 Rule Testing

## **8.7 Ethical and Legal Considerations in Rule Generation**

The process of automated rule development raises ethical and legal concerns, specifically with the accuracy of rules and the possibility of false positives. The dependence of integration on automated processes highlights the necessity for ongoing validation, refinement, and monitoring. Ethical considerations comprise the imperative to ensure that the regulations established adhere to regulatory frameworks and do not violate privacy rights.

### **Conclusion:**

The chapter delves into the process of rule creation and automation workflow, showcasing the integrated system's capacity for innovation. The connection greatly enhances the process of rule construction by utilising OpenVAS reports, Python scripts, and user-input customization. This streamlined approach empowers organisations to efficiently identify vulnerabilities and promptly address new threats.

## Chapter 9: Analysis and Results

This chapter delves deeper into the integrated system's efficacy, efficiency, and impact on the security of the network. By exposing the system to several attack scenarios, the chapter measures the system's ability to detect intrusions and assesses its contribution to proactive defence.

### 9.1 Evaluation of Intrusion Detection

To determine the effectiveness of the integrated system, a series of intrusion detection experiments are conducted. To simulate attacks against experimental environments, identical simulations of genuine threat vectors are used. Focusing on false positives, false negatives, and response times, the system's ability to detect and respond to these attacks accurately is evaluated.

The results cast light on the system's responsiveness and durability.

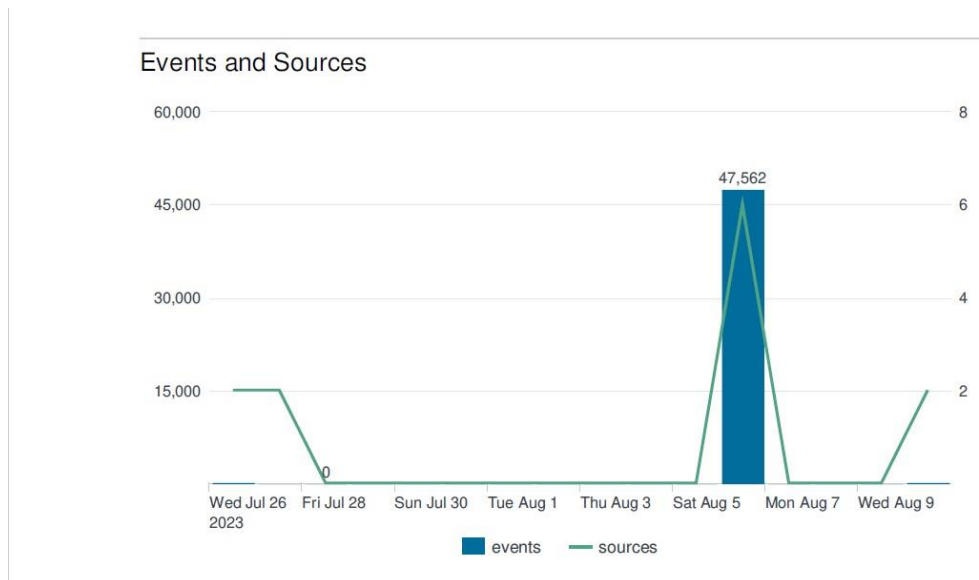


Figure 23 Event and Sources

### 9.2 Detection of Vulnerabilities

The examination of the vulnerability assessment component within the integration is conducted employing complete scans of the testing environment. The efficacy of OpenVAS in the identification and prioritization of vulnerabilities is assessed, hence providing insights into its capacity to accurately detect and highlight potential problems. The investigation highlights the significance of integration in facilitating the generation of practical insights for enhancing system security and managing software updates.

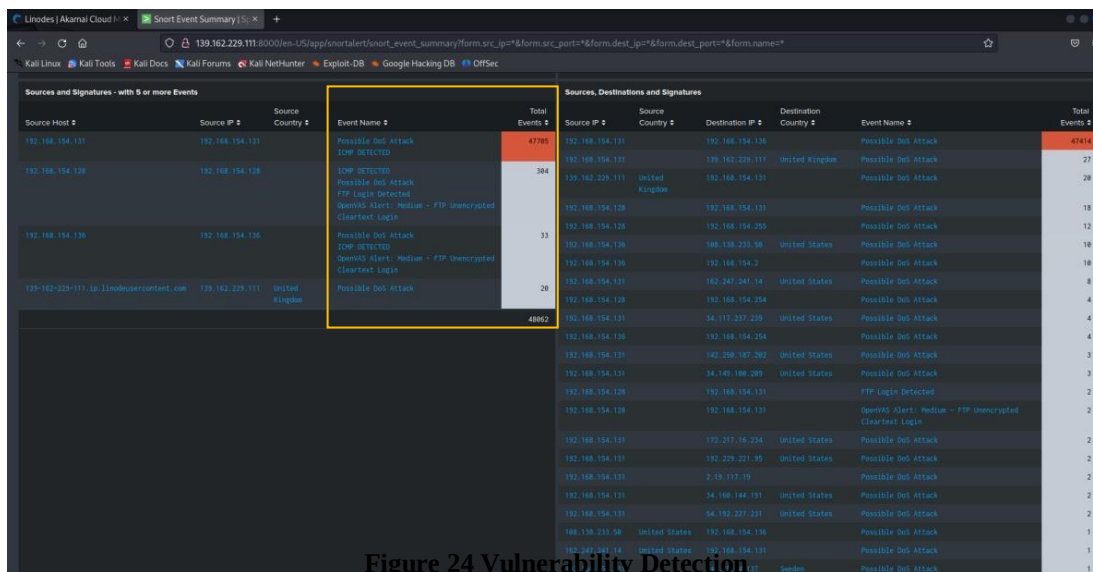


Figure 24 Vulnerability Detection

### 9.3 User-Input-Based Rule Evaluation

The evaluation of the script for the customization of rules, depending on user input, is carried out with a primary emphasis on its adaptability and effectiveness. Security managers possess the capability to customise rules according to unique scenarios and threat profiles through the manipulation of numerous factors. The assessment of the produced rules is carried out using a sequence of attack simulations, showcasing the influence of the alteration on the accuracy of threat identification.

### 9.4 Splunk Visualization and Real-Time Monitoring

The contribution of Splunk to real-time monitoring and data visualisation is a crucial component of this investigation. The assessment focuses on the integration's capacity to consolidate logs, establish connections between events, and offer practical insights. Splunk dashboards provide a visual depiction of the network's security posture by illustrating real-time threat patterns, attack origins, and top vulnerabilities.

## **9.5 Ethical, Legal, and Societal Implications**

The process of integration requires an examination of its ethical, legal, and societal implications. The utilization of automated rule creation and network traffic manipulation gives rise to ethical considerations related to the protection of data privacy, potential harm, and the necessity for consent. The evaluation of the compatibility of integration with legal frameworks and its influence on society is conducted to guarantee ethical functioning within wider contexts.

## **9.6 Performance Metrics and Comparative Analysis**

The performance of the integrated system is evaluated using key characteristics such as detection rate, false positive rate, and response time. The integration's efficacy is explained in context by comparing these KPIs to industry standards and best practices. The comparative analysis provides a benchmark against which the integration's performance can be measured.

## **9.7 Future Prospects and Scalability**

Finally, this chapter discusses the system's future potential and scalability. The integration's future development, expansion to wider network environments, and response to emerging threats are all explored. The analysis' findings give a road map for improving the integration and realizing its full potential in protecting digital ecosystems.

## **Conclusion:**

This chapter's study and results provide a concrete picture of the integrated system's contributions to network security. This chapter provides actual proof of the integration's practicability and potential for proactive defence by examining intrusion detection, vulnerability identification, rule customization, Splunk's visualization capabilities, and ethical issues.

## **Chapter 10: Discussion and Implications**

This chapter examines the integration's outcomes, limitations, contributions to the field, and alignment with larger industry trends in depth. This chapter adds to the topic of the integration's significance by delving into the implications for network security, ethical concerns, and the integration's impact on cybersecurity practises.

### **10.1 Contributions to Network Security**

The integration of Snort, OpenVAS, and Splunk makes a major contribution to network security. The automated rule generation method speeds up the development of intrusion detection rules, allowing organizations to respond to emerging threats more quickly. The synergy of the integration enables organizations to take a proactive defence posture that aligns with modern cybersecurity principles.

### **10.2 Ethical Dimensions and Privacy Considerations**

The functional features of the integration pose ethical concerns about data privacy, user permission, and potential harm. Handling network traffic, which may contain sensitive information, is required for automated rule development. The need to protect user privacy, acquiring appropriate permissions, and minimizing potential harm cannot be emphasized. This chapter addresses these ethical considerations thoroughly and emphasizes the importance of balancing security and privacy.

### **10.3 Alignment with Industry Trends**

The alignment of the integration with current industry trends is critical. The integration's mindset aligns with the industry's shift towards automation, machine learning, and holistic cybersecurity methods. The integration represents a forward-thinking strategy for dealing with modern cyber threats by automating rule formulation, integrating vulnerability assessment, and employing log management for real-time information.

#### **10.4 Limitations and Future Enhancements**

To create transparency and set realistic expectations, it is critical to identify the integration's limitations. Scalability, adaptability to varied network contexts, and the possibility of false positives in automated rule creation all require consideration. Future improvements could include improving rule correctness, using machine learning methods, and resolving potential performance bottlenecks.

#### **10.5 Implications for Cybersecurity Practices**

The implications of the integration of cybersecurity practices go beyond technological elements. It emphasizes the importance of proactive defence and ongoing monitoring in dealing with evolving threats. The ability of the integration to centralize, correlate, and visualize data emphasizes the importance of real-time insights in enhancing incident response and threat detection capabilities.

#### **10.6 Ethical Guidelines and Compliance**

The implementation of integration to ethical standards and regulatory frameworks is of utmost importance in effectively going through the complex ethical landscape. Ensuring adherence to data protection laws, industry standards, and privacy legislation is of utmost importance. This chapter examines the ethical compliance of integration and its role in promoting acceptable cybersecurity practices.

#### **10.7 Integration's Place in the Ecosystem**

This discussion's objective is to illustrate the function of integration within the larger cybersecurity ecosystem. Due to its ability to improve intrusion detection, vulnerability assessment, and real-time monitoring, this technology must be incorporated into every comprehensive security plan. By incorporating multiple tools and aligning with prevalent industry trends, businesses can increase their resistance to constantly evolving cyber threats.

**Conclusion:**

This chapter elaborates on the integration's implications, ethical considerations, alignment with industry trends, and potential improvements. This chapter contributes to the discussion surrounding the significance of the integration by explaining its contributions to network security, addressing its limitations, and considering its broader impact on cybersecurity practices.



## Chapter 11: Conclusion and Future Directions

The final section summarizes the research findings, highlighting the integration's contributions, ramifications, and possibilities for future advancements. This chapter provides closure while encouraging future network security endeavors by summarizing the important insights, reflecting on the study experience, and proposing prospects for additional exploration.

### 11.1 Summary of Key Findings

A concise review of the research's primary findings summarises the integration's accomplishments. Building a strong network defence with Snort, OpenVAS, and Splunk offers a vast amount of data. When these technologies integrate, they produce a well-organized defensive system that not only detects threats but also allows for proactive prevention and informed decision-making.

The problems that signature-based and anomaly-based detection methodologies in intrusion detection systems (IDS) encounter have been investigated. This comprehension served as the foundation for our in-depth examination of Snort, an open-source intrusion detection and prevention system that has developed into an essential component of our integrated security ecosystem.

Our investigation expanded to include OpenVAS, where we discovered its value as a robust vulnerability assessment tool. Its ability to detect and alert administrators to potential network gaps allows administrators to rectify vulnerabilities and improve network resilience.

Splunk, a data analysis and visualisation software that serves as the security apparatus's eyes and ears, was also on our list of stops. Its combination with Snort enhances our event detection capabilities, while real-time analysis provides administrators with information on how to analyse, mitigate, and finally eradicate any attacks.

Our journey into automated rule creation was marked by a combination of innovative thinking and practicality. Individual judgment and technical efficacy are symbiotically linked in the Python processes I built. These scripts highlight the balance between automation and human intelligence by converting vulnerability reports into Snort-compatible rules and enabling userinput-based customization.

In search of an unbeatable defence, we examined our security system's legal, social, ethical, and professional components. This investigation shed light on the broader implications of human action, emphasizing the significance of balancing technology advancements with ethical considerations.

As we near the end of our project, we have gained a better understanding of the cybersecurity landscape. Aside from technology, our holistic defence architecture produces a symphony of focus, ability, and strategic understanding. Snort, OpenVAS, and Splunk work together to build an environment where attacks are recognized, vulnerabilities are patched, and insights drive us forward. This journey exemplifies our concept by demonstrating the intersection of knowledge, technology, and resilience with the goal of a safer digital society.

## **11.2 Reflection on Research**

This chapter pauses for reflection, reflecting the research's path, problems, and discoveries. The conclusion of our study journey provides a time for reflection, an opportunity to reflect on the paths we've taken and the insights we've gained. The journey towards cybersecurity has been an array of discoveries, obstacles, and growth, aided by the integration of Snort, OpenVAS, and Splunk.

The various signature-based and anomaly-based detection approaches revealed the unique aspects of threat identification. This understanding established the framework for our in-depth exploration of Snort, demonstrating its architecture, rule-based detection, and position as a guardian guarding against digital invasions.

The spotlight was then cast on OpenVAS, a beacon of vulnerability assessment that allows administrators to map and conquer possible vulnerabilities. Our journey through its installation, configuration, and use revealed its ability to harden networks by addressing weaknesses in advance. Splunk emerged as our data analysis and visualisation guide, providing us with real-time insights on network activities and intrusions. The combination of Snort's detection abilities and Splunk's ability to analyse revealed a new realm of network defence, where understanding guides to action. Our investigation into automated rule generation demonstrated the power of innovation. Our Python scripts turned everyday tasks into a symphony of automation and personalization. The interaction of technology scripts and human inputs highlighted the seamless integration of cutting-edge capabilities and smart decisions.

This journey, however, was not just technical; it also included ethical, social, legal, and professional aspects. The investigation of these dimensions raised our awareness of the wider implications of

our technological endeavors, reminding us that responsible innovation is founded on ethical considerations.

This research journey has been more than just a search for knowledge; it has been a profound experience. We have gone beyond theory, embracing the delicate combination of concept and practice, automation, and human insight. The combination of Snort, OpenVAS, and Splunk is more than just a technological effort; it represents a combination of invention, insight, and resilience.

We take a step back and reflect on the miles we've traversed, grateful for the insights gained, the challenges faced, and the progress we've attained. This project journey emphasizes that the pursuit of cybersecurity is a collective one—one in which information pushes us ahead, aided by technology and directed by ethical ideals.

### 11.3 Avenues for Future Exploration

The chapter focuses on the horizon, pointing to potential future directions. In addition to highlighting the complex framework of integrated cybersecurity using Snort, OpenVAS, and Splunk, our research journey prepares the way for exciting new avenues of investigation. These prospective paths drive us to delve deeper, innovate further, and enhance our defence mechanisms as technology and threats transform.

1. **Machine Learning Integration:** The combination of machine learning and cybersecurity has enormous potential. Future studies could look towards incorporating machine learning algorithms to improve threat detection accuracy, adapt to developing attack techniques, and reduce false positives.
2. **Behavioral Analysis:** Beyond signatures, the behavioural analysis could provide insights into unusual user behaviors, hence improving detection capabilities. Investigating user and object behavior analytics could add a new layer of threat detection.
3. **IoT Security:** It is vital to protect these endpoints as the Internet of Things (IoT) evolves. Investigating how we might strengthen our integrated architecture to protect IoT devices from potential threats could be an exciting experiment.
4. **Automation Refinement:** More complicated and context-aware rules could be produced by improving the automated components of rule creation. This could include integrating

threat information feeds and allowing rules to be dynamically changed in response to emerging risks.

5. **Privacy and Compliance:** As data privacy requirements become more strict research could focus on how our integrated system can meet compliance standards such as GDPR. The problem of ensuring that the integration respects user privacy while improving security is motivating.
6. **Real-time Threat Hunting:** Future research could include building real-time threat hunting strategies based on Splunk's capabilities. Instead of depending exclusively on automated alerts, administrators would be able to actively seek for possible risks.

#### 11.4 Final Reflections

The chapter finishes with last thoughts on the integration's significance, potential, and broader influence. The successful integration of Snort, OpenVAS, and Splunk represents the cybersecurity intent of collaboration, automation, and complete defence. The integration is a milestone in reinforcing digital environments against emerging cyber challenges, demonstrating creativity, research, and practicality.

## **Chapter 12: Recommendations and Practical Insights**

This chapter offers professionals, organizations, and academics attempting to take advantage of the combination of Snort, OpenVAS, and Splunk for increased network security a set of recommendations and practical insights gained from the research path.

### **12.1 Implementation Guidelines**

A detailed set of instructions is provided earlier to facilitate the integration's successful implementation. This document details the installation and configuration of Snort, OpenVAS, and Splunk, as well as their coordination and data flow integration. These recommendations are enhanced by insights into user-input-based rule customization, rule administration, and cloud system integration.

### **12.2 Ethical Considerations and Compliance**

This section addresses the maintenance of ethical standards and regulatory compliance in the context of the ethical considerations that accompany the integration. Obtaining user consent, assuring data anonymization, and adhering to data protection regulations are emphasized. To effectively navigate the ethical landscape, recommendations on data management, log retention policies, and incident reporting are provided.

### **12.3 Continuous Monitoring and Adaptation**

Continuous monitoring and adaptation are essential to the integration's success. The author provides practical advice on establishing a proactive defence posture through real-time monitoring of Snort alerts, OpenVAS reports, and Splunk dashboards. It is recommended to utilize automated alerts, incident response processes, and regular vulnerability assessments to maintain a resilient security posture.

## **12.4 Training and Skill Enhancement**

Implementation of the integration requires personnel competent in navigating its complexities. This section supports cybersecurity professional training and skill enhancement programs. Suggestions for workshops, certifications, and hands-on training sessions are provided to enable practitioners to effectively utilize the integration's capabilities.

## **12.5 Collaboration and Knowledge Sharing**

This chapter emphasizes the significance of cybersecurity community engagement and knowledge exchange. It is advised to build networked teams, participate in cybersecurity forums, and contribute to open-source projects. Collaboration ensures continual integration improvement, the ability to respond to emerging risks, and resistance against evolving attack vectors.

## **Conclusion:**

This chapter's guidelines and practical insights provide a road map for professionals, organizations, and researchers to maximize the benefits of integration. This chapter links the gap between theory and practice by providing concrete suggestions on implementation, ethical considerations, continual monitoring, skill enhancement, collaboration, and future research possibilities.

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## Appendix A:

### A. Snort Rule Example:

```
alert tcp any any -> $HOME_NET 80 (msg:"Possible HTTP Malware Download";  
content:"/malware.exe"; nocase; sid:100001;)
```

### B. OpenVAS original report:

IP,Hostname,Port,Port Protocol,CVSS,Severity,Solution Type,NVT Name,Summary,Specific Result,NVT OID,CVEs,Task ID,Task Name,Timestamp,Result ID,Impact,Solution,Affected Software/OS,Vulnerability Insight,Vulnerability Detection Method,Product Detection Result,BIDs,CERTs,Other References

192.168.154.131,,21,tcp,6.4,Medium,"Mitigation","Anonymous FTP Login Reporting","Reports if the remote FTP Server allows anonymous logins.", "It was possible to login to the remote FTP service with the following anonymous account(s):

anonymous:anonymous@example.com ftp:anonymous@example.com

",1.3.6.1.4.1.25623.1.0.900600,"CVE-1999-0497",e3af86fa-8809-467f-89aa-6de3bbcd45ac,"kali1",2023-08-06T11:06:45Z,762f71ea-3d52-4224-9bd2-04b64e8a0365,"Based on the files accessible via this anonymous FTP login and the permissions of this account an attacker might be able to:

- gain access to sensitive files

- upload or delete files.", "If you do not want to share files, you should disable anonymous logins.", "", "A host that provides an FTP service may additionally provide

Anonymous FTP access as well. Under this arrangement, users do not strictly need an account on the host. Instead the user typically enters 'anonymous' or 'ftp' when prompted for username. Although users are commonly asked to send their email address as their password, little to no verification is actually performed on the supplied data.

Remark: NIST don't see 'configuration issues' as software flaws so the referenced CVE has a severity of 0.0. The severity of this VT has been raised by Greenbone to still report a configuration issue on the target.", " Details:

Anonymous FTP Login Reporting

(OID: 1.3.6.1.4.1.25623.1.0.900600)

Version used: 2021-10-20T09:03:29Z

", "", "", "", ""

192.168.154.131,,21,tcp,4.8,Medium,"Mitigation","FTP Unencrypted Cleartext Login","The remote host is running a FTP service that allows cleartext logins over unencrypted connections.", "The remote FTP service accepts logins without a previous sent 'AUTH TLS' command. Response(s):

Non-anonymous sessions: 331 Please specify the password.

Anonymous sessions: 331 Please specify the password.

",1.3.6.1.4.1.25623.1.0.108528,"",e3af86fa-8809-467f-89aa-6de3bbcd45ac,"kali1",2023-0806T11:06:45Z,ef8abe2e-ab84-43ee-b657-6a05915cd4a0,"An attacker can uncover login names and passwords by sniffing traffic to the

FTP service.", "Enable FTPS or enforce the connection via the 'AUTH TLS' command. Please see the manual of the FTP service for more information.", "", "", "Tries to login to a non FTPS enabled FTP service without sending a

'AUTH TLS' command first and checks if the service is accepting the login without enforcing the use of the 'AUTH TLS' command.

Details:

FTP Unencrypted Cleartext Login

(OID: 1.3.6.1.4.1.25623.1.0.108528)

Version used: 2023-07-14T16:09:27Z

", "", "", "", ""

### **C. Report converted by Python script:**

IP: 192.168.154.131 Hostname:

Port: 21

Protocol: tcp

CVSS: 6.4

Severity: Medium

NVT Name: Anonymous FTP Login Reporting

Summary: Reports if the remote FTP Server allows anonymous logins.

Specific Result: It was possible to login to the remote FTP service with the following anonymous account(s):

anonymous:anonymous@example.com ftp:anonymous@example.com

NVT OID: 1.3.6.1.4.1.25623.1.0.900600

CVEs: CVE-1999-0497

Task ID: e3af86fa-8809-467f-89aa-6de3bbcd45ac

Task Name: kali1

Timestamp: 2023-08-06T11:06:45Z

Result ID: 762f71ea-3d52-4224-9bd2-04b64e8a0365

Impact: Based on the files accessible via this anonymous FTP login and the permissions of this account an attacker might be able to:

- gain access to sensitive files

- upload or delete files.

Solution: If you do not want to share files, you should disable anonymous logins.

Affected Software/OS:

Vulnerability Insight: A host that provides an FTP service may additionally provide Anonymous FTP access as well. Under this arrangement, users do not strictly need an account on the host. Instead the user typically enters 'anonymous' or 'ftp' when prompted for username. Although users are commonly asked to send their email address as their password, little to no verification is actually performed on the supplied data.

Remark: NIST don't see 'configuration issues' as software flaws so the referenced CVE has a severity of 0.0. The severity of this VT has been raised by Greenbone to still report a configuration issue on the target.

Vulnerability Detection Method:

Details:

Anonymous FTP Login Reporting

(OID: 1.3.6.1.4.1.25623.1.0.900600)

Version used: 2021-10-20T09:03:29Z

Product Detection Result:

BIDs:

CERTs:

Other References:

----- IP:

192.168.154.131 Hostname:

Port: 21

Protocol: tcp

CVSS: 4.8

Severity: Medium

NVT Name: FTP Unencrypted Cleartext Login

Summary: The remote host is running a FTP service that allows cleartext logins over unencrypted connections.

Specific Result: The remote FTP service accepts logins without a previous sent 'AUTH TLS' command. Response(s):

Non-anonymous sessions: 331 Please specify the password.

Anonymous sessions: 331 Please specify the password.

NVT OID: 1.3.6.1.4.1.25623.1.0.108528 CVEs:

Task ID: e3af86fa-8809-467f-89aa-6de3bbcd45ac

Task Name: kali1

Timestamp: 2023-08-06T11:06:45Z

Result ID: ef8abe2e-ab84-43ee-b657-6a05915cd4a0

Impact: An attacker can uncover login names and passwords by sniffing traffic to the FTP service.

Solution: Enable FTPS or enforce the connection via the 'AUTH TLS' command. Please see the manual of the FTP service for more information.

Affected Software/OS:

Vulnerability Insight:

Vulnerability Detection Method: Tries to login to a non FTPS enabled FTP service without sending a

'AUTH TLS' command first and checks if the service is accepting the login without enforcing the use of the 'AUTH TLS' command.

Details:

FTP Unencrypted Cleartext Login

(OID: 1.3.6.1.4.1.25623.1.0.108528)

Version used: 2023-07-14T16:09:27Z

Product Detection Result:

BIDs:

CERTs:

Other References:

#### D. Snort Automated Rule

- alert tcp any any -> any 21 (msg:"OpenVAS Alert: Medium - Anonymous FTP Login Reporting"; content:"USER CVE-1999-0497"; sid:100000;)
- alert tcp any any -> any 21 (msg:"OpenVAS Alert: Medium - FTP Unencrypted Cleartext Login"; content:"USER "; sid:100001;)
- alert tcp any any -> any (msg:"OpenVAS Alert: High - Operating System (OS) End of Life (EOL) Detection"; content:"USER "; sid:100000;)
- alert tcp any any -> any 8787 (msg:"OpenVAS Alert: High - Distributed Ruby (dRuby/DRb) Multiple Remote Code Execution Vulnerabilities"; content:"USER "; sid:100001;)
- alert tcp any any -> any 1524 (msg:"OpenVAS Alert: High - Possible Backdoor: Ingreslock"; content:"USER "; sid:100002;)
- alert tcp any any -> any 80 (msg:"OpenVAS Alert: High - TWiki XSS and Command Execution Vulnerabilities"; content:"USER CVE-2008-5304,CVE-2008-5305"; sid:100003;)
- alert tcp any any -> any 513 (msg:"OpenVAS Alert: High - rlogin Passwordless Login"; content:"USER "; sid:100004;)
- alert tcp any any -> any 512 (msg:"OpenVAS Alert: High - The rexec service is running"; content:"USER CVE-1999-0618"; sid:100005;)
- alert tcp any any -> any 8009 (msg:"OpenVAS Alert: High - Apache Tomcat AJP RCE Vulnerability (Ghostcat)"; content:"USER CVE-2020-1938"; sid:100006;)
- alert tcp any any -> any 3632 (msg:"OpenVAS Alert: High - DistCC RCE Vulnerability (CVE-2004-2687)"; content:"USER CVE-2004-2687"; sid:100007;)
- alert tcp any any -> any 5432 (msg:"OpenVAS Alert: High - PostgreSQL Default Credentials (PostgreSQL Protocol)"; content:"USER "; sid:100008;)
- alert tcp any any -> any 5900 (msg:"OpenVAS Alert: High - VNC Brute Force Login"; content:"USER "; sid:100009;)
- alert tcp any any -> any 6697 (msg:"OpenVAS Alert: High - UnrealIRCd Authentication Spoofing Vulnerability"; content:"USER CVE-2016-7144"; sid:100010;)
- alert tcp any any -> any 3306 (msg:"OpenVAS Alert: High - MySQL / MariaDB Default Credentials (MySQL Protocol)"; content:"USER CVE-2001-0645,CVE-2004-2357,CVE2006-1451,CVE-2007-2554,CVE-2007-6081,CVE-2009-0919,CVE-2014-3419,CVE-2015-4669,CVE-2016-6531,CVE-2018-15719"; sid:100011;)

- alert tcp any any -> any 80 (msg:"OpenVAS Alert: High - PHP-CGI-based setups vulnerability when parsing query string parameters from php files."; content:"USER CVE-2012-1823,CVE-2012-2311,CVE-2012-2336,CVE-2012-2335"; sid:100012;)
- alert tcp any any -> any 1099 (msg:"OpenVAS Alert: High - Java RMI Server Insecure Default Configuration RCE Vulnerability"; content:"USER CVE-2011-3556"; sid:100013;)
- alert tcp any any -> any 6697 (msg:"OpenVAS Alert: High - UnrealIRCd Backdoor"; content:"USER CVE-2010-2075"; sid:100014;)
- alert tcp any any -> any 6200 (msg:"OpenVAS Alert: High - vsftpd Compromised Source Packages Backdoor Vulnerability"; content:"USER "; sid:100015;)
- alert tcp any any -> any 21 (msg:"OpenVAS Alert: High - vsftpd Compromised Source Packages Backdoor Vulnerability"; content:"USER "; sid:100016;)
- alert tcp any any -> any 514 (msg:"OpenVAS Alert: High - rsh Unencrypted Cleartext Login"; content:"USER CVE-1999-0651"; sid:100017;)
- alert tcp any any -> any 80 (msg:"OpenVAS Alert: High - phpinfo() output Reporting"; content:"USER "; sid:100018;)
- alert tcp any any -> any 80 (msg:"OpenVAS Alert: High - Test HTTP dangerous methods"; content:"USER "; sid:100019;)
- alert tcp any any -> any 513 (msg:"OpenVAS Alert: High - The rlogin service is running"; content:"USER CVE-1999-0651"; sid:100020;)
- alert tcp any any -> any 21 (msg:"OpenVAS Alert: High - FTP Brute Force Logins Reporting"; content:"USER CVE-1999-0501,CVE-1999-0502,CVE-1999-0507,CVE-1999-0508,CVE-2018-19063,CVE-2018-19064"; sid:100021;)
- alert tcp any any -> any 2121 (msg:"OpenVAS Alert: High - FTP Brute Force Logins Reporting"; content:"USER CVE-1999-0501,CVE-1999-0502,CVE-1999-0507,CVE-1999-0508,CVE-2018-19063,CVE-2018-19064"; sid:100022;)
- alert tcp any any -> any 5432 (msg:"OpenVAS Alert: High - SSL/TLS: OpenSSL CCS Man in the Middle Security Bypass Vulnerability"; content:"USER CVE-2014-0224"; sid:100023;)
- alert tcp any any -> any 25 (msg:"OpenVAS Alert: Medium - Multiple Vendors STARTTLS Implementation Plaintext Arbitrary Command Injection Vulnerability"; content:"USER CVE-2011-0411,CVE-2011-1430,CVE-2011-1431,CVE-2011-1432,CVE-2011-1506,CVE-2011-1575,CVE-2011-1926,CVE-2011-2165"; sid:100024;)



- alert tcp any any -> any 80 (msg:"OpenVAS Alert: Medium - TWiki Cross-Site Request Forgery Vulnerability - Sep10"; content:"USER CVE-2009-4898"; sid:100025;)
- alert tcp any any -> any 21 (msg:"OpenVAS Alert: Medium - Anonymous FTP Login Reporting"; content:"USER CVE-1999-0497"; sid:100026;)
- alert tcp any any -> any 80 (msg:"OpenVAS Alert: Medium - jQuery < 1.9.0 XSS Vulnerability"; content:"USER CVE-2012-6708"; sid:100027;)
- alert tcp any any -> any 80 (msg:"OpenVAS Alert: Medium - TWiki < 6.1.0 XSS Vulnerability"; content:"USER CVE-2018-20212"; sid:100028;)

## Appendix B:

### A. Snort Usage

```
sudo snort -q -l /var/log/snort -i eth0 -A fast -c /etc/snort/snort.conf
```

**q** – Quite mode **i** – Network interface **l** – Log directory **A** –

**Fast alert c** – Configuration file

